

ENGINEERING CHEMISTRY LAB

Engineering Chemistry Lab is a laboratory course that is designed to provide students with practical experience in the application of chemical principles to engineering problems. The course typically includes experiments that demonstrate the principles of chemical reactions, thermodynamics, electrochemistry, and materials science.

Some of the key objectives of the Engineering Chemistry Lab are: - To provide students with hands-on experience in conducting experiments and analyzing data. - To develop students' skills in problem-solving, critical thinking, and teamwork. - To reinforce the concepts learned in the lecture component of the Engineering Chemistry course. - To introduce students to the use of laboratory equipment and techniques.

Typical experiments that may be performed in an Engineering Chemistry Lab include: - Synthesis of compounds and materials. - Analysis of chemical reactions and reaction rates. - Measurement of thermodynamic properties such as enthalpy and entropy. - Investigation of electrochemical processes such as corrosion and battery operation. - Characterization of materials using techniques such as X-ray diffraction and scanning electron microscopy.

In summary, the Engineering Chemistry Lab is an essential component of an engineering education, providing students with the opportunity to apply their knowledge of chemistry to real-world problems and to develop important skills that will serve them well in their future careers.

Course Objectives:

1. To provide a comprehensive understanding of the subject matter.
2. To develop critical thinking and analytical skills.
3. To enhance problem-solving abilities.
4. To encourage independent learning and self-motivation.
5. To foster effective communication and collaboration skills.
6. To promote the application of theoretical knowledge to real-world scenarios.
7. To prepare students for further academic or professional pursuits in the field.

1. To enable the students to understand about the fundamental concepts of analytical instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are essential tools in various fields such as chemistry, biology, medicine, and environmental science. Some of the fundamental concepts of analytical instruments include:

- **Accuracy and precision:** Accuracy refers to how close a measurement

is to the true value, while precision refers to how close repeated measurements are to each other. Both are important in analytical measurements.

- **Sensitivity:** Sensitivity refers to the ability of an instrument to detect small changes in the quantity being measured. A highly sensitive instrument can detect very small changes in the sample.
- **Selectivity:** Selectivity refers to the ability of an instrument to distinguish between different components in a sample. A highly selective instrument can accurately measure the quantity of a specific component in the presence of other components.
- **Calibration:** Calibration is the process of adjusting an instrument to ensure that its measurements are accurate. This is typically done by comparing the instrument's measurements to those of a known standard.
- **Signal-to-noise ratio:** The signal-to-noise ratio (SNR) is a measure of the quality of a measurement. It is the ratio of the signal (the quantity being measured) to the noise (unwanted variations in the measurement). A high SNR indicates a high-quality measurement.

These are some of the fundamental concepts of analytical instruments that students should understand. Understanding these concepts will enable students to use analytical instruments effectively and accurately in their work.

2. To enable the students to understand about the analysis of chloride content, hardness, alkalinity of water.

- **Chloride content:** Chloride is one of the major anions found in water and wastewater. The analysis of chloride content in water is important because high levels of chloride can be harmful to aquatic life and can also affect the taste of drinking water. The most common method for analyzing chloride content in water is the argentometric method, which involves titrating a water sample with a silver nitrate solution.
- **Hardness:** Hardness in water is caused by the presence of dissolved calcium and magnesium ions. It is important to analyze the hardness of water because it can affect the performance of water-using appliances and can also cause scaling in pipes and boilers. The most common method for analyzing water hardness is the EDTA titration method.
- **Alkalinity:** Alkalinity is a measure of the water's ability to neutralize acids. It is important to analyze the alkalinity of water because it can affect the pH of the water and can also influence the effectiveness of water treatment processes. The most common method for analyzing water alkalinity is the titration method using a standard acid solution.

These are some of the key points that students should understand about the analysis of chloride content, hardness, and alkalinity of water. It is important

for students to have a good understanding of these concepts in order to properly analyze and interpret water quality data.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of a liquid.

- **pH** is a measure of the acidity or basicity of a liquid. It is defined as the negative logarithm of the hydrogen ion concentration in the solution. The pH scale ranges from 0 to 14, with 7 being neutral. A pH less than 7 indicates an acidic solution, while a pH greater than 7 indicates a basic solution.
- **Surface tension** is the property of a liquid that allows it to resist an external force. It is caused by the cohesive forces between the liquid molecules. Surface tension can be measured using various methods, such as the drop weight method, the maximum bubble pressure method, and the Wilhelmy plate method.
- **Viscosity** is the measure of a fluid's resistance to flow. It is commonly referred to as the thickness of a fluid. Viscosity can be measured using various methods, such as the capillary tube method, the falling ball method, and the rotational viscometer method.

These properties of a liquid are important in various fields, such as chemistry, physics, and engineering. Understanding how to measure and interpret these properties can provide valuable insights into the behavior of liquids.

4. To enable the students to understand about the preparation of different resins.

Resins are solid or highly viscous substances that are typically convertible into polymers. They are usually mixtures of organic compounds, mainly terpenes and their derivatives. Resins are used in a variety of applications, including adhesives, coatings, and inks.

The preparation of resins involves several steps, including:

1. **Selection of raw materials:** The raw materials used in the preparation of resins vary depending on the type of resin being produced. Common raw materials include natural and synthetic monomers, solvents, and catalysts.
2. **Polymerization:** Polymerization is the process of chemically bonding monomers together to form a polymer. This can be achieved through various methods, including addition, condensation, and ring-opening polymerization.
3. **Purification:** After polymerization, the resulting resin may contain impurities that need to be removed. This can be achieved through various methods, including filtration, distillation, and extraction.

4. **Formulation:** Once the resin has been purified, it can be formulated to meet specific application requirements. This may involve the addition of additives, such as plasticizers, stabilizers, and pigments.
5. **Processing:** The final step in the preparation of resins is processing, which involves shaping the resin into its final form. This can be achieved through various methods, including molding, extrusion, and casting.

In summary, the preparation of resins involves the selection of raw materials, polymerization, purification, formulation, and processing. Each of these steps is critical to producing a high-quality resin that meets the desired specifications.

5. To enable the students to understand about the synthesis of organic compounds such as adipic acid and paracetamol by conventional and green route.

- Adipic acid is an important organic compound that is used in the production of nylon. It can be synthesized through a conventional route, which involves the oxidation of cyclohexanol or cyclohexanone with nitric acid. This process produces large amounts of nitrous oxide, a greenhouse gas that contributes to climate change.
- A green route for the synthesis of adipic acid involves the use of hydrogen peroxide as an oxidizing agent, which produces water as the only byproduct. This process is more environmentally friendly and sustainable.
- Paracetamol, also known as acetaminophen, is a widely used pain reliever and fever reducer. It can be synthesized through a conventional route, which involves the nitration of phenol followed by the reduction of the resulting nitro compound.
- A green route for the synthesis of paracetamol involves the use of enzymes, such as peroxidases, to catalyze the reaction. This process is more environmentally friendly and produces fewer byproducts.
- Green chemistry is an approach to chemical synthesis that aims to reduce the environmental impact of chemical processes. It involves the use of sustainable and renewable resources, the reduction of waste and the use of safer and less toxic chemicals.
- The synthesis of organic compounds such as adipic acid and paracetamol through green routes is an example of how green chemistry can be applied to reduce the environmental impact of chemical processes. It is important for students to understand the principles of green chemistry and how it can be applied in the synthesis of organic compounds.

LIST OF EXPERIMENTS

1. **Milgram Experiment:** A study on obedience to authority figures, conducted by Stanley Milgram in the 1960s.

2. **Stanford Prison Experiment:** A study of the psychological effects of becoming a prisoner or prison guard, conducted by Philip Zimbardo in 1971.
3. **Pavlov's Dogs:** A study on classical conditioning, conducted by Ivan Pavlov in the 1890s.
4. **Asch Conformity Experiment:** A study on conformity, conducted by Solomon Asch in the 1950s.
5. **Hawthorne Effect:** A study on the effect of changes in working conditions on productivity, conducted by Elton Mayo in the 1920s.
6. **Bobo Doll Experiment:** A study on the effects of observed aggression, conducted by Albert Bandura in the 1960s.
7. **Little Albert Experiment:** A study on classical conditioning of fear, conducted by John B. Watson and Rosalie Rayner in 1920.
8. **The Marshmallow Test:** A study on delayed gratification, conducted by Walter Mischel in the 1960s.
9. **The Trolley Problem:** A thought experiment in ethics, first introduced by Philippa Foot in 1967.
10. **The Ultimatum Game:** A study on economic behavior, first introduced by Werner Güth, Rolf Schmittberger, and Bernd Schwarze in 1982.

1. Calibration of Analytical Equipment and Apparatus

Calibration is the process of verifying the accuracy of an instrument or apparatus by comparing its measurements with a known standard. This is important for ensuring that the results obtained from the instrument are accurate and reliable.

Here are some key points to consider when calibrating analytical equipment and apparatus:

- Calibration should be performed regularly to ensure that the instrument is functioning properly and providing accurate results.
- The frequency of calibration will depend on factors such as the type of instrument, its usage, and the environment in which it is used.
- Calibration should be performed using a standard that is traceable to a recognized national or international standard.
- The calibration process should be documented, including the date of calibration, the standard used, and the results obtained.
- Any adjustments or corrections made to the instrument during calibration should also be documented.
- If the instrument is found to be out of calibration, appropriate corrective action should be taken, such as adjusting the instrument or replacing it if necessary.

In summary, calibration is an essential step in ensuring the accuracy and reliability of analytical equipment and apparatus. It should be performed regularly and documented to ensure that the instrument is functioning properly and providing accurate results.

2. Determination of Hardness of water sample by EDTA method

- Hardness of water is determined by the presence of calcium and magnesium ions.
- EDTA (ethylenediaminetetraacetic acid) is a chelating agent that can bind to these ions and form a complex.
- In the EDTA method, a solution of EDTA is added to a water sample until all the calcium and magnesium ions have formed complexes with the EDTA.
- The amount of EDTA used is then measured, which can be used to calculate the hardness of the water sample.
- This method is commonly used in laboratories to determine the hardness of water samples.

3. Determination of Alkalinity of water sample

Alkalinity is a measure of the ability of water to neutralize acids. It is an important parameter in water quality analysis, as it provides information about the buffering capacity of water.

1. Alkalinity is determined by titrating a water sample with a standard acid solution, usually sulfuric acid or hydrochloric acid, to a specific pH endpoint.
2. The endpoint is usually determined by a color change of an indicator, such as phenolphthalein or methyl orange.
3. The amount of acid required to reach the endpoint is used to calculate the alkalinity of the water sample.
4. Alkalinity is typically reported in milligrams per liter (mg/L) as calcium carbonate (CaCO_3).
5. Alkalinity can also be determined by using an automatic titrator or by measuring the pH of the water sample before and after the addition of a known amount of acid.

4. Determination of pH by titrimetric method

Titrimetric method is a technique used to determine the pH of a solution. This method involves the use of a titrant, which is a solution of known concentration, to react with the solution being tested. The following are the steps involved in the determination of pH by titrimetric method:

1. **Preparation of the titrant:** The titrant is prepared by dissolving a known amount of a strong acid or base in a known volume of water. The concentration of the titrant is then calculated.
2. **Selection of an appropriate indicator:** An indicator is a substance that changes color at a specific pH range. The indicator is chosen based on the pH range of the solution being tested.

3. **Titration:** The solution being tested is placed in a flask, and the indicator is added. The titrant is then added dropwise from a burette while stirring the solution. The addition of the titrant is continued until the indicator changes color, indicating that the endpoint of the titration has been reached.
4. **Calculation of pH:** The volume of the titrant used is recorded, and the number of moles of the titrant is calculated. The number of moles of the acid or base in the solution being tested is then calculated using the stoichiometry of the reaction. The pH of the solution is then calculated using the concentration of the acid or base.

This method is accurate and precise, and it is commonly used in laboratories to determine the pH of solutions. However, it requires the use of specialized equipment and chemicals, and it may not be suitable for all applications.

5. Determination of surface tension of given liquid

Surface tension is a property of liquids that arises due to the cohesive forces between the molecules at the surface. It is defined as the force acting per unit length perpendicular to the line of separation of the liquid from another phase. There are several methods to determine the surface tension of a given liquid, including:

1. **Maximum Bubble Pressure Method:** This method involves measuring the maximum pressure required to blow a bubble of air through a liquid. The surface tension is calculated using the Laplace equation.
2. **Drop Weight Method:** This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube. The surface tension is calculated using the weight of the droplet and the radius of the tube.
3. **Wilhelmy Plate Method:** This method involves immersing a thin flat plate into a liquid and measuring the force required to detach it from the liquid. The surface tension is calculated using the force and the perimeter of the plate.
4. **Du Nouy Ring Method:** This method involves immersing a thin ring into a liquid and measuring the force required to detach it from the liquid. The surface tension is calculated using the force and the circumference of the ring.
5. **Capillary Rise Method:** This method involves measuring the height to which a liquid rises in a thin tube due to capillary action. The surface tension is calculated using the height of the liquid, the radius of the tube, and the density of the liquid.

Each of these methods has its own advantages and limitations, and the choice of method depends on the specific requirements of the experiment. It is important

to carefully calibrate the instruments and control the experimental conditions to obtain accurate results.

6. Determination of Viscosity of a given liquid by viscometer

Viscosity is a measure of a fluid's resistance to flow. It is commonly measured using a device called a viscometer. Here are the steps to determine the viscosity of a given liquid using a viscometer:

1. **Select the appropriate viscometer:** Different types of viscometers are available for different types of liquids and measurement ranges. Choose the one that is suitable for the liquid you want to measure.
2. **Calibrate the viscometer:** Before taking any measurements, it is important to calibrate the viscometer to ensure accurate results. Follow the manufacturer's instructions for calibration.
3. **Prepare the sample:** The liquid to be measured should be free of air bubbles and at a consistent temperature. Pour the liquid into the viscometer, taking care not to introduce any air bubbles.
4. **Take the measurement:** Follow the manufacturer's instructions for taking the measurement. This typically involves timing the flow of the liquid through a narrow tube or capillary.
5. **Calculate the viscosity:** Use the data obtained from the measurement and the calibration to calculate the viscosity of the liquid. The calculation method will vary depending on the type of viscometer used.
6. **Record and report the results:** Record the results of the measurement, including the temperature and any other relevant information. Report the results in the appropriate units, such as centipoise (cP) or Pascal-seconds (Pa.s).

It is important to follow the manufacturer's instructions and good laboratory practices when using a viscometer to ensure accurate and reliable results.

7. Determination of the strength of Ferrous ammonium sulfate using external indicator

1. Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate.
2. Its chemical formula is $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$.
3. It is commonly used as a standard solution in redox titrations due to its stability and the ease of preparation.
4. The strength of a ferrous ammonium sulfate solution can be determined by titrating it against a standard solution of potassium permanganate (KMnO_4) using an external indicator.
5. The external indicator used in this titration is potassium ferricyanide ($\text{K}_3\text{Fe}(\text{CN})_6$).

6. The reaction between ferrous ammonium sulfate and potassium permanganate is a redox reaction, where potassium permanganate is reduced to manganese sulfate (MnSO_4) and ferrous ammonium sulfate is oxidized to ferric ammonium sulfate ($(\text{NH}_4)_3\text{Fe}(\text{SO}_4)_3$).
7. The end point of the titration is indicated by the disappearance of the blue color of the potassium ferricyanide solution, which indicates the complete oxidation of the ferrous ions to ferric ions.
8. The strength of the ferrous ammonium sulfate solution can be calculated using the volume of potassium permanganate solution used and the molarity of the potassium permanganate solution.

8. Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is an oxidizing agent that can be used to determine the strength of a substance.
2. An internal indicator is used to determine the endpoint of the titration.
3. The internal indicator changes color when the reaction is complete, indicating that the titration is finished.
4. The strength of the potassium dichromate can then be calculated using the volume of the titrant and the concentration of the substance being titrated.
5. This method is commonly used in analytical chemistry to determine the strength of a substance.

9. Determination of available chlorine in bleaching powder

Bleaching powder, also known as calcium hypochlorite, is a chemical compound with the formula $\text{Ca}(\text{OCl})_2$. It is commonly used as a disinfectant and bleaching agent. The available chlorine in bleaching powder refers to the amount of chlorine that can be released as a disinfectant when the powder is dissolved in water.

The available chlorine in bleaching powder can be determined using the following steps:

1. Weigh a sample of bleaching powder and dissolve it in water to form a solution.
2. Add excess potassium iodide (KI) to the solution. The iodide ions react with the available chlorine to form iodine.
3. Titrate the iodine formed with a standard solution of sodium thiosulfate ($\text{Na}_2\text{S}_2\text{O}_3$) using starch as an indicator. The reaction between iodine and thiosulfate is as follows: $2\text{S}_2\text{O}_3^{2-} + \text{I}_2 \rightarrow \text{S}_4\text{O}_6^{2-} + 2\text{I}^-$
4. From the volume of thiosulfate solution used, the amount of iodine formed can be calculated. From this, the amount of available chlorine in the bleaching powder can be determined.

This method is known as iodometric titration and is commonly used to determine the available chlorine in bleaching powder. It is a simple and accurate method for determining the available chlorine content of bleaching powder.

10. Determination of chloride content in water sample

1. Chloride content in water can be determined using a titration method known as the Mohr method.
2. The Mohr method involves titrating the water sample with a silver nitrate solution.
3. A chromate indicator is used to signal the endpoint of the titration.
4. The silver nitrate reacts with the chloride ions in the water sample to form a white precipitate of silver chloride.
5. The chromate ions react with the excess silver ions to form a red precipitate of silver chromate, indicating the endpoint of the titration.
6. The volume of silver nitrate solution used in the titration is used to calculate the concentration of chloride ions in the water sample.
7. The Mohr method is a simple and accurate method for determining the chloride content in water samples.
8. Other methods for determining the chloride content in water samples include ion chromatography and ion-selective electrodes.
9. It is important to determine the chloride content in water samples as high levels of chloride can have negative impacts on aquatic life and can also affect the taste and odor of drinking water.
10. Regular monitoring of the chloride content in water sources is necessary to ensure the safety and quality of the water supply.

11. Preparation of Phenol formaldehyde (PF) resin

Phenol formaldehyde (PF) resin, also known as phenolic resin, is a synthetic polymer obtained by the reaction of phenol with formaldehyde. Here are the steps involved in the preparation of PF resin:

1. **Phenol and formaldehyde** are mixed in a reactor in the presence of an **acid catalyst** such as hydrochloric acid or sulfuric acid.
2. The mixture is heated to a temperature of around **90°C** and stirred continuously.
3. The reaction between phenol and formaldehyde produces **water** as a byproduct, which is removed by distillation.
4. The reaction continues until the desired degree of polymerization is achieved. The degree of polymerization can be controlled by adjusting the ratio of phenol to formaldehyde, the reaction temperature, and the reaction time.
5. The resulting resin is then cooled and solidified. It can be further processed into various forms such as powders, granules, or preforms.

PF resin is widely used in the production of various products such as laminates,

adhesives, and molded objects. It is known for its high strength, heat resistance, and chemical resistance.

12. Preparation of Urea formaldehyde (UF) resin

Urea-formaldehyde (UF) resin, also known as urea-methanal, is a non-transparent thermosetting resin or plastic made from urea and formaldehyde when they are heated in the presence of a mild base such as ammonia or pyridine.

The most common method of preparation for commercial UF resin is the addition of a second amount of urea during the reaction. The ratio of urea to formaldehyde is between 1:2 and 1:2.2 and therefore methylation can take place at in a short amount of time at temperatures between 90 and 95 °C, with a mixture being maintained under reflux.

The synthesis of all UF resins is a typical three-step procedure (alkaline-acidic-alkaline). First, formaldehyde is poured into a three-necked flask and adjusted to pH 7.5 to pH 8.0 with 20% sodium hydroxide solution. The first amount of urea is added to give the F/U molar ratio of 2.0.

Urea is manufactured from carbon dioxide and ammonia at a temperature of 135–200 °C and at a pressure of 70–230 atmospheres. Formaldehyde is manufactured by the oxidation of methanol which can be produced from the reaction of carbon dioxide with hydrogen or can be derived from petroleum.

13. Preparation of Adipic acid / Paracetamol

Adipic acid and paracetamol are two different compounds with different chemical structures and properties. Here is a brief overview of their preparation methods:

Adipic Acid

Adipic acid is an organic compound with the formula $(\text{CH}_2)_4(\text{COOH})_2$. It is a white crystalline powder that is mainly used in the production of nylon. Adipic acid can be prepared by several methods, including:

1. Oxidation of cyclohexanol or cyclohexanone with nitric acid. This is the most common industrial method for the production of adipic acid.
2. Oxidation of cyclohexene with hydrogen peroxide in the presence of a tungsten catalyst.
3. Hydrogenation of phenol to cyclohexanone, followed by oxidation with nitric acid.

Paracetamol

Paracetamol, also known as acetaminophen, is an analgesic and antipyretic drug that is widely used to relieve pain and reduce fever. It is prepared by the following method:

1. Nitration of phenol to produce para-nitrophenol.
2. Reduction of para-nitrophenol to para-aminophenol.
3. Acetylation of para-aminophenol with acetic anhydride to produce paracetamol.

14. Determination of Cell Conductance of a solution

Cell conductance is a measure of a solution's ability to conduct an electric current. It is determined by the concentration and mobility of ions in the solution. Here are the steps to determine the cell conductance of a solution:

1. Prepare the solution to be tested by dissolving a known amount of solute in a known volume of solvent.
2. Fill the conductivity cell with the solution to be tested.
3. Connect the conductivity cell to a conductivity meter.
4. Turn on the conductivity meter and allow it to stabilize.
5. Record the conductivity reading displayed on the meter.
6. Calculate the cell conductance of the solution using the formula: $G = k \times C$, where G is the cell conductance, k is the cell constant, and C is the conductivity reading.

It is important to note that the cell constant, k , is specific to the conductivity cell being used and must be determined experimentally or provided by the manufacturer. Additionally, the temperature of the solution can affect its conductivity, so it is important to measure and record the temperature of the solution during the experiment.

15. Determination of Rate constant of hydrolysis of esters

1. The hydrolysis reaction of an ester in pure water is a slow reaction.
2. When a mineral acid like hydrochloric acid is added, the rate of the reaction is enhanced since the H^+ ions from the mineral acid act as the catalyst.
3. The hydrolysis of esters is catalyzed by either an acid or a base.
4. Acidic hydrolysis is simply the reverse of esterification.
5. The ester is heated with a large excess of water containing a strong-acid catalyst.
6. Like esterification, the reaction is reversible and does not go to completion.
7. The rate constant for the hydrolysis reaction of an ester by dilute acid is $0.6931 \times 10^{-3} \text{ s}^{-1}$.
8. The rate constant after calculation from the graphs was approximately $0.003 \text{ min}^{-1} \text{ cm}^{-3}$ for the 1ml and 2ml of ethyl acetate.
9. When ethyl ethanoate is heated under reflux with a dilute acid such as dilute hydrochloric acid or dilute sulfuric acid, the ester reacts with the water present to produce ethanoic acid and ethanol.

16. Element detection and identification of functional groups in organic compounds

1. **Element detection** refers to the process of identifying the presence of specific elements in a given sample.
2. **Functional groups** are specific groups of atoms within molecules that are responsible for the characteristic chemical reactions of those molecules.
3. The identification of functional groups in organic compounds is important for understanding their chemical properties and reactivity.
4. Common methods for element detection and identification of functional groups in organic compounds include:
 - **Infrared spectroscopy (IR)**: This technique measures the absorption of infrared radiation by the sample, which can provide information about the functional groups present.
 - **Nuclear magnetic resonance spectroscopy (NMR)**: This technique measures the interaction of nuclear spins with an applied magnetic field, providing information about the chemical environment of the nuclei and the functional groups present.
 - **Mass spectrometry (MS)**: This technique measures the mass-to-charge ratio of ions produced from the sample, providing information about the molecular weight and structure of the compound, including the functional groups present.
 - **Elemental analysis**: This technique measures the relative amounts of different elements in the sample, providing information about the elemental composition of the compound.
5. These techniques can be used in combination to provide a more complete picture of the chemical composition and structure of the sample.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose 10 experiments from the list provided.
- The instructor also has the option to change any two of the experiments from the list.
- This allows the instructor to tailor the experiments to the needs and interests of the students.
- It is important for students to carefully read and understand the instructions provided by the instructor.
- Students should also be prepared for any changes to the list of experiments.

Course Outcomes:

- Define the objectives and goals of the course.
- Identify the skills and knowledge that students are expected to acquire by the end of the course.
- Describe the methods of assessment used to evaluate student learning.

- Outline the course content and the sequence of topics to be covered.
- Specify the instructional materials and resources required for the course.
- Discuss the teaching strategies and techniques used to facilitate student learning.
- Explain how the course outcomes align with the program outcomes and institutional goals.
- Provide a clear and concise summary of the course outcomes for students to refer to throughout the course.

Upon completion of the course the student should be able to:

1. Demonstrate a thorough understanding of the course material.
2. Apply the concepts and theories learned in the course to real-world situations.
3. Analyze and evaluate information critically and effectively.
4. Communicate ideas and arguments clearly and effectively in both written and oral forms.
5. Work collaboratively with others to achieve common goals.
6. Demonstrate ethical and professional behavior in all aspects of the course.
7. Use technology effectively to enhance learning and communication.
8. Engage in continuous learning and professional development.

Course Outcomes Bloom's

Course outcomes are statements that describe the knowledge, skills, and abilities that students should have acquired by the end of a course. These outcomes are typically aligned with the course objectives and are used to assess the effectiveness of the course in achieving its goals.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. It is commonly used to design course outcomes and assessments. The taxonomy consists of six levels, arranged in a hierarchy from lower-order thinking skills to higher-order thinking skills:

1. **Remembering:** The ability to recall or retrieve previously learned information.
2. **Understanding:** The ability to comprehend the meaning of material.
3. **Applying:** The ability to use learned material in new and concrete situations.
4. **Analyzing:** The ability to break down material into its component parts so that its organizational structure may be understood.
5. **Evaluating:** The ability to make judgments about the value of ideas or materials.
6. **Creating:** The ability to put parts together to form a new whole.

Course outcomes should be written in a way that reflects the level of thinking required by the course. For example, a course outcome for a lower-level course

might focus on remembering and understanding, while a course outcome for a higher-level course might focus on analyzing and evaluating.

Level

- A level is a tool used to determine if a surface is horizontal (level) or vertical (plumb).
- Levels can be used in construction, carpentry, surveying, and many other applications.
- There are several types of levels, including spirit levels, laser levels, and water levels.
- Spirit levels use a liquid-filled vial with an air bubble to indicate levelness.
- Laser levels project a laser beam to create a straight line on a surface.
- Water levels use the principle that water will always find its own level to determine levelness.
- Levels can vary in size and accuracy, with larger and more precise levels being used for more demanding applications.
- It is important to use a level when installing or building anything that needs to be level or plumb, such as shelves, cabinets, or walls.
- Using a level can help ensure that a project is completed accurately and to a high standard.

CO-1 Get an understanding of the use of different analytical instruments. K3

Analytical instruments are used to measure physical, chemical, and biological properties of samples. These instruments are used in a variety of fields, including chemistry, biology, physics, and engineering. Some common analytical instruments include:

1. **Spectrophotometers:** These instruments measure the absorption or transmission of light by a sample. They are commonly used to determine the concentration of a substance in a solution.
2. **Chromatographs:** These instruments separate mixtures into their individual components. They are commonly used to identify and quantify the components of a mixture.
3. **Mass spectrometers:** These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify the chemical composition of a sample.
4. **Electrochemical instruments:** These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.
5. **Microscopes:** These instruments magnify small objects to make them visible to the naked eye. They are commonly used to study the structure of cells and tissues.

Each of these instruments has its own unique capabilities and limitations. Understanding the use of different analytical instruments is essential for selecting the appropriate instrument for a given task and for interpreting the results obtained from the instrument.

CO-2 Measure the molecular / system properties such as surface tension

Surface tension is a physical property of liquids that results from the cohesive forces between the molecules at the surface of a liquid. It is defined as the force acting per unit length on an imaginary line drawn on the liquid surface, tending to pull the surface molecules together.

Some key points to remember about surface tension are:

1. Surface tension is caused by the cohesive forces between the molecules at the surface of a liquid.
2. It is a measure of the energy required to increase the surface area of a liquid.
3. Surface tension is typically measured in units of force per unit length, such as N/m or dyn/cm.
4. The surface tension of a liquid can be affected by various factors, including temperature, the presence of surfactants, and the nature of the liquid itself.
5. There are several methods for measuring surface tension, including the drop weight method, the maximum bubble pressure method, and the Wilhelmy plate method.

Viscosity, Conductance of Solution, Chloride and Iron Content in the Water

1. **Viscosity** is a measure of a fluid's resistance to flow. It is determined by the internal friction of the fluid, which is caused by the cohesive forces between the molecules. The viscosity of formation water is a function of pressure, temperature, and dissolved solids. In general, brine viscosity increases with increasing pressure, increasing salinity, and decreasing temperature.
2. **Conductance of solution** refers to the ability of a solution to conduct an electric current. The conductivity of water is affected by the presence of dissolved ions, such as chloride ions. The more chloride ions present in the water, the higher the conductivity. In general, the conductivity of waterways in the United States varies from 50 to 1500 $\mu\text{mhos/cm}$, and inland freshwater lake studies reveal a conductivity of 150 to 500 $\mu\text{mhos/cm}$. High-quality deionized water has a conductivity of about 0.05 S/cm at 25 °C, typical drinking water is in the range of 200–800 S/cm, while seawater is about 50 mS/cm (or 0.05 S/cm).
3. **Chloride content** in water refers to the concentration of chloride ions

in the water. Chloride ions are approximately 1.9 percent of the mass of seawater.

4. **Iron content** in water refers to the concentration of iron ions in the water. Iron is an essential element for all forms of life, but it can also be a contaminant in water. High levels of iron in water can cause discoloration, staining, and an unpleasant taste.

K3

- K3 is a term that can refer to several different things, depending on the context in which it is used.
- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a visa category for spouses of U.S. citizens.
- In mathematics, K3 could refer to a complete graph with three vertices.
- In music, K3 is a Belgian-Dutch girl group.
- Without more context, it is difficult to determine which specific meaning of K3 is being referred to.

CO-3 Measure the hardness and alkalinity of the water. K3

Hardness and alkalinity are two important parameters that are used to measure the quality of water. Hardness refers to the concentration of calcium and magnesium ions in the water, while alkalinity refers to the water's ability to neutralize acids.

Here are some points to consider when measuring the hardness and alkalinity of water:

1. Hardness can be measured using a titration method, where a solution of known concentration is added to the water sample until a color change occurs. The amount of solution required to cause the color change is used to calculate the hardness of the water.
2. Alkalinity can be measured using a similar titration method, where a solution of known concentration is added to the water sample until a pH change occurs. The amount of solution required to cause the pH change is used to calculate the alkalinity of the water.
3. There are also test kits available that can be used to measure the hardness and alkalinity of water. These kits typically contain reagents that react with the water sample to produce a color change, which can be compared to a color chart to determine the hardness or alkalinity of the water.
4. It is important to measure the hardness and alkalinity of water regularly, as changes in these parameters can indicate changes in the water quality. High levels of hardness or alkalinity can cause problems such as scaling or

corrosion, while low levels can indicate the presence of pollutants or other contaminants.

5. When measuring the hardness and alkalinity of water, it is important to follow the instructions provided with the test kit or titration method carefully to ensure accurate results. It is also important to use clean, uncontaminated equipment and to handle the water sample carefully to avoid introducing contaminants.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an organic compound that is also known as carbolic acid. It is a weak acid that mostly forms phenoxide ions by dropping one positive hydrogen ion (H^+) from the hydroxyl group. There are several methods for the preparation of phenol, including:

1. From Haloarenes: Chlorobenzene is an example of a haloarene which is formed by the monosubstitution of the benzene ring. When chlorobenzene is fused with sodium hydroxide at 623K and 320 atm, sodium phenoxide is produced .
2. From Benzene Sulphonic Acid: At 573 – 623K, sodium phenoxide is produced by fusing the sodium salt of benzene sulphonic acid with sodium hydroxide. Sodium phenoxide is converted to phenol when it is treated with dilute acid or carbon dioxide.
3. From Coal Tar: Phenol is extracted from the middle oil fraction for commercial purposes at 443 – 503K.
4. Raschig's Process: The reaction of benzene, hydrogen chloride and air at 523K.
5. From Cumene: Phenol can be prepared in large quantities by the air oxidation of cumene.
6. From Diazonium Salts: Phenol can be prepared in small amounts by the hydrolysis of diazonium salts.
7. From Phenylboronic Acid: Phenol can be prepared in small amounts by the peroxide oxidation of phenylboronic acid.

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

- **Formaldehyde** is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.
- **Urea-formaldehyde resin (UF)** is synthesized using formaldehyde and urea as raw materials by the “alkali-acid-alkali” method . UF is a non-transparent thermosetting resin or polymer.

- UF is used in adhesives, plywood, particle board, medium-density fibre-board (MDF), and molded objects.
- Unlike most thermosetting resins, UF often shows the appearance of crystal domains .
- Melamine, polyvinyl alcohol, and adipic acid dihydrazide can be added to UF resin as modifiers for reducing the free formaldehyde.

K3

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CO-5 Estimate the rate constant of reaction. K3

The rate constant of a reaction, denoted by k , is a measure of the speed at which a chemical reaction occurs. It is defined as the proportionality constant between the rate of the reaction and the concentrations of the reactants raised to their respective powers, as given by the rate law.

To estimate the rate constant of a reaction, the following steps can be followed:

1. Determine the order of the reaction with respect to each reactant by performing experiments in which the concentration of one reactant is varied while the others are kept constant.
2. Write the rate law for the reaction using the determined orders of reaction.
3. Measure the initial rate of the reaction for several sets of initial concentrations of the reactants.
4. Substitute the initial concentrations and initial rates into the rate law to obtain a set of simultaneous equations.
5. Solve the simultaneous equations to obtain the value of the rate constant, k .

It is important to note that the rate constant is temperature-dependent and its value will change with changes in temperature. The relationship between the rate constant and temperature is given by the Arrhenius equation.

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- I'm sorry, but I don't have enough information to write about the topic "Page 29 of 40".

- Could you please provide more context or details about the topic you want me to write about?
- Is “Page 29 of 40” a title or a reference to a specific document or book?
- Providing more information will help me generate a more accurate and informative response.

ENGINEERING PHYSICS LAB KCS

Engineering Physics Lab KCS is a laboratory course that is designed to provide students with hands-on experience in the application of physics principles to engineering problems. The course typically covers topics such as:

1. Mechanics: Students will learn about the fundamental principles of mechanics, including Newton’s laws of motion, conservation of energy and momentum, and rotational dynamics.
2. Electricity and Magnetism: Students will learn about the fundamental principles of electricity and magnetism, including Coulomb’s law, Gauss’s law, and Faraday’s law of induction.
3. Optics: Students will learn about the fundamental principles of optics, including reflection, refraction, and diffraction.
4. Thermodynamics: Students will learn about the fundamental principles of thermodynamics, including the first and second laws of thermodynamics, and the concept of entropy.
5. Modern Physics: Students will learn about the fundamental principles of modern physics, including the theory of relativity and quantum mechanics.

The course is designed to provide students with a strong foundation in the fundamental principles of physics, and to prepare them for further study in engineering and related fields. Students will have the opportunity to work on experiments and projects that will help them to develop their problem-solving and critical thinking skills. The course is typically taken by students in their first or second year of an engineering program.

Course Objectives

- To provide students with a comprehensive understanding of the subject matter.
- To equip students with the necessary skills and knowledge to apply the concepts learned in real-life situations.
- To develop critical thinking and problem-solving skills.
- To encourage independent learning and self-directed study.
- To prepare students for further academic study or professional development in the field.

- To foster an appreciation for the subject and its relevance to various aspects of life.

Fundamental Concepts of Analytical Instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are essential tools in various fields, including chemistry, biology, medicine, and environmental science. Here are some fundamental concepts of analytical instruments:

1. **Accuracy and Precision:** Accuracy refers to how close a measurement is to the true value, while precision refers to how close repeated measurements are to each other. Analytical instruments must be both accurate and precise to provide reliable results.
2. **Sensitivity:** Sensitivity refers to the ability of an instrument to detect small changes in the property being measured. A highly sensitive instrument can detect even minute changes, providing more detailed and accurate results.
3. **Calibration:** Calibration is the process of adjusting an instrument to ensure that it provides accurate and precise measurements. This is typically done by comparing the instrument's readings to those of a known standard.
4. **Signal-to-Noise Ratio:** The signal-to-noise ratio (SNR) is a measure of the quality of a measurement. It compares the strength of the signal (the property being measured) to the background noise (unwanted signals that can interfere with the measurement). A high SNR indicates a high-quality measurement.
5. **Resolution:** Resolution refers to the smallest change in the property being measured that can be detected by the instrument. A high-resolution instrument can detect very small changes, providing more detailed and accurate results.

These are some of the fundamental concepts of analytical instruments that students should understand. By understanding these concepts, students can better appreciate the capabilities and limitations of analytical instruments and use them more effectively in their work.

Analysis of Chloride Content, Hardness, and Alkalinity

1. **Chloride Content:** Chloride content refers to the amount of chloride ions present in a sample of water. Chloride ions are negatively charged

particles that can be found in many types of salts, including sodium chloride (common table salt). The presence of chloride ions in water can indicate contamination from sources such as seawater intrusion, industrial waste, or road salt runoff.

2. **Hardness:** Water hardness is a measure of the concentration of calcium and magnesium ions in water. These ions can cause problems in industrial and domestic settings by forming deposits on pipes and fixtures, interfering with the action of soap and detergents, and reducing the efficiency of water heaters. Hardness is typically measured in grains per gallon (gpg) or milligrams per liter (mg/L) as calcium carbonate.
3. **Alkalinity:** Alkalinity is a measure of the ability of water to neutralize acids. It is primarily determined by the presence of carbonate, bicarbonate, and hydroxide ions in water. Alkalinity is important in maintaining a stable pH in water and preventing corrosion of pipes and fixtures. It is typically measured in milligrams per liter (mg/L) as calcium carbonate.

These three parameters are important in the analysis of water quality and can provide valuable information about the sources of contamination and the suitability of water for various uses.

Water

Water is a transparent, tasteless, odorless, and nearly colorless chemical substance, which is the main constituent of Earth's streams, lakes, and oceans, and the fluids of most living organisms. It is vital for all known forms of life, even though it provides no calories or organic nutrients.

- Water is a liquid at standard temperature and pressure.
- Water covers 71% of the Earth's surface.
- Water is essential for the survival of all known forms of life.
- Water is a polar molecule, meaning it has a partial positive charge near the hydrogen atoms and a partial negative charge near the oxygen atom.
- Water has a high heat capacity, meaning it can absorb a lot of heat before its temperature rises.
- Water has a high surface tension, allowing it to form droplets and to support small objects on its surface.
- Water is a universal solvent, meaning it can dissolve many substances.
- Water is involved in many chemical reactions, including hydrolysis and hydration.
- Water exists in three states: solid (ice), liquid, and gas (water vapor).
- Water can exist in all three states at the same time in a state known as the triple point.

Measure of pH, Surface Tension and Viscosity

1. **pH:** pH is a measure of the acidity or basicity of a solution. It is defined as the negative logarithm of the hydrogen ion concentration. A pH of 7 is considered neutral, while a pH less than 7 is acidic and a pH greater than 7 is basic.
2. **Surface Tension:** Surface tension is the property of a liquid that allows it to resist an external force. It is caused by the cohesive forces between the liquid molecules. Surface tension is measured in units of force per unit length, typically in millinewtons per meter (mN/m).
3. **Viscosity:** Viscosity is the measure of a fluid's resistance to flow. It is commonly referred to as the thickness of a fluid. Viscosity is measured in units of dynamic viscosity, typically in pascal-seconds ($\text{Pa} \cdot \text{s}$) or poise (P).

These three properties are important in various fields, including chemistry, physics, and engineering. Understanding these properties can help in the design and analysis of various systems and processes.

Liquid

- A liquid is one of the four fundamental states of matter, along with solid, gas, and plasma.
- It is characterized by its ability to conform to the shape of its container while retaining a constant volume.
- Liquids have a definite volume, but no definite shape.
- The molecules in a liquid are in constant motion and are loosely packed, allowing them to slide past one another.
- The attractive forces between the molecules in a liquid are strong enough to keep the molecules close together, but not strong enough to keep them in a fixed position.
- The temperature at which a substance changes from a solid to a liquid is called its melting point, and the temperature at which it changes from a liquid to a gas is called its boiling point.
- Liquids can be classified as either Newtonian or non-Newtonian, depending on their flow behavior.
- A Newtonian liquid has a constant viscosity, meaning its resistance to flow does not change with the applied shear stress.
- A non-Newtonian liquid, on the other hand, has a variable viscosity, meaning its resistance to flow changes with the applied shear stress.
- Examples of liquids include water, milk, blood, and oil.

Preparation of Different Resins

Resins can be derived from various sources such as polymers, monomers, oils, fatty acids, and natural sources. They have a wide range of applications in industries such as paints and coatings, personal care and home care, pharmaceuticals, rubber, plastics, electrical and jewelry articles, decorative items, and wood coatings .

There are two categories of resinous products: natural resins and prepared resins . The preparation of resins can vary depending on the type of resin being produced. For example, by manipulating the structure of the linker, resins ranging from extremely acid labile to base labile can be prepared. Using linkers additionally allows the preparation of resins with special applications such as DHP resin utilized as a solid-phase support for alcohols or Weinreb resin utilized for preparing aldehydes and ketones .

Some common types of resins include phenolic resins, alkyd resins, epoxy resins, UV resins, polyester resins, and polyurethane resins. The majority of resins are made up of two elements, namely the base resin and a catalyst (which is a hardener). Combining the two elements results in a chemical reaction that allows the resin to set .

Synthesis of Adipic Acid

Adipic acid is a substance used for the preparation of nylon, polyurethanes, and other products of commercial interest. The industrial synthesis of adipic acid uses nitric acid oxidation of a mixture of cyclohexanone and cyclohexanol termed “KA oil” (ketone and alcohol-oil).

1. Traditional feedstock for the synthesis of adipic acid is benzene. Benzene is hydrogenated using Ni-Al₂O₃ catalyst at 370–800 psi to cyclohexane.
2. Cyclohexane is then oxidized using Co catalyst to a mixture of cyclohexanone and cyclohexanol.
3. This mixture is then converted to adipic acid using ammonium vanadate and nitric acid, with the help of a catalyst.

This procedure generates nitrous oxide, a substance that increases greenhouse gases, damages the ozone layer, and causes acid rain.

Acid and Paracetamol by Conventional and Green Route

- An alternative route for the synthesis of N-acetyl-p-aminophenol (paracetamol, PAR) and acetylsalicylic acid (aspirin, ASA) using principles of green chemistry has been proposed. This involves solvent-free synthesis combined with microwave-energy irradiation.

- In 2014, Joncour et al., established a green approach involving direct amidation of hydroquinone to obtain paracetamol with maximum conversion. This eliminates the problems associated with the earlier techniques such as, the extra step involved, unwanted ortho- isomerization, generation of the sulfate salt.
- Acetaminophen can be synthesized from phenol in three steps. In this synthetic route, the solvent from step two is kept to help maximize atom economy. The first step is an electrophilic aromatic substitution on phenol with nitric acid to create p-nitrophenol.

List of Experiments

1. **The Milgram Experiment:** A series of social psychology experiments conducted by Stanley Milgram in the 1960s, which measured the willingness of study participants to obey an authority figure who instructed them to perform acts that conflicted with their personal conscience.
2. **The Stanford Prison Experiment:** A social psychology experiment conducted by Philip Zimbardo in 1971, which aimed to investigate the psychological effects of perceived power, focusing on the struggle between prisoners and prison officers.
3. **The Asch Conformity Experiment:** A series of studies conducted by Solomon Asch in the 1950s, which demonstrated the power of conformity in groups and the influence of majority opinion on individual judgment.
4. **The Pavlov's Dog Experiment:** A series of experiments conducted by Ivan Pavlov in the 1890s, which demonstrated the principle of classical conditioning, where a neutral stimulus can be used to elicit a conditioned response.
5. **The Marshmallow Test:** A study on delayed gratification conducted by Walter Mischel in the 1960s and 1970s, where children were offered a choice between a small immediate reward or a larger reward if they waited for a short period.

Calibration of Analytical Equipment and Apparatus

Calibration is the process of verifying the accuracy of a measuring instrument by comparing its readings with those of a known standard. This is important for ensuring the reliability and accuracy of analytical results.

1. Calibration is necessary for all types of analytical equipment and apparatus, including balances, pipettes, thermometers, and spectrophotometers.
2. The frequency of calibration depends on the type of equipment, its usage, and the requirements of the relevant regulatory bodies.
3. Calibration can be performed using certified reference materials, which are

materials with known properties that have been certified by a recognized authority.

4. Calibration can also be performed using external standards, which are materials with known properties that have been prepared in-house or obtained from an external source.
5. Calibration should be performed according to a documented procedure, and the results should be recorded and retained for future reference.
6. Calibration should be performed by trained personnel, and the equipment should be maintained and operated according to the manufacturer's instructions.
7. If the equipment is found to be out of calibration, it should be adjusted or repaired, and the calibration should be repeated until the equipment is within the acceptable range.

2. Determination of Hardness of water sample by EDTA method

- Hardness of water is a measure of the concentration of calcium and magnesium ions in water.
- The EDTA (ethylenediaminetetraacetic acid) method is a common method used to determine the hardness of water.
- In this method, a solution of EDTA is used to titrate a water sample.
- The EDTA binds with the calcium and magnesium ions present in the water, forming a complex.
- An indicator such as Eriochrome Black T is used to signal the end point of the titration.
- The amount of EDTA used in the titration is proportional to the concentration of calcium and magnesium ions in the water sample.
- The hardness of the water sample can be calculated using the volume of EDTA solution used in the titration.
- This method is simple, accurate, and widely used in the determination of water hardness.

Determination of Alkalinity of Water Sample

Alkalinity is a measure of the ability of water to neutralize acids. It is an important parameter in water quality analysis, as it provides information about the buffering capacity of water. Here are the steps to determine the alkalinity of a water sample:

1. Collect a water sample in a clean container.
2. Measure the pH of the sample using a pH meter or pH test strips.
3. If the pH is above 4.5, titrate the sample with a standard solution of sulfuric acid (H_2SO_4) or hydrochloric acid (HCl) using a burette.

4. Record the volume of acid required to lower the pH to 4.5.
5. Calculate the alkalinity of the sample using the formula: $\text{Alkalinity (mg/L as CaCO}_3) = (\text{Volume of acid used} \times \text{Normality of acid} \times 50,000) / \text{Volume of sample}$.

This method is known as the titrimetric method and is commonly used for the determination of alkalinity in water samples. It is important to note that the results obtained using this method are dependent on the accuracy of the pH measurement and the normality of the acid used for titration. Therefore, it is important to calibrate the pH meter and standardize the acid solution before performing the analysis.

Determination of pH by Titrimetric Method

1. Titration is a laboratory technique used to determine the concentration of an unknown solution by reacting it with a solution of known concentration.
2. The pH of a solution can be determined by titrating it with a strong acid or base.
3. The titration curve is a graph of pH versus volume of titrant added. The equivalence point is the point at which the amount of titrant added is chemically equivalent to the amount of analyte present.
4. The pH at the equivalence point depends on the strength of the acid or base being titrated. For a strong acid-strong base titration, the pH at the equivalence point is 7. For a weak acid-strong base titration, the pH at the equivalence point is greater than 7. For a strong acid-weak base titration, the pH at the equivalence point is less than 7.
5. Indicators are used to visually signal the end point of the titration. The indicator should change color at a pH close to the equivalence point.
6. The pH can be calculated using the Henderson-Hasselbalch equation for weak acid-strong base and strong acid-weak base titrations.
7. The accuracy of the pH determination depends on the precision of the titration and the accuracy of the standard solution used.

Determination of Surface Tension of Given Liquid

Surface tension is a property of liquids that arises due to the cohesive forces between the molecules at the surface. It is defined as the force acting per unit length on the surface of a liquid. There are several methods to determine the surface tension of a given liquid, including:

1. **Capillary rise method:** This method involves measuring the height of the liquid that rises in a capillary tube due to the surface tension of the liquid. The surface tension is then calculated using the formula: $\text{Surface tension} = \frac{(\rho g h r)}{2}$, where ρ is the density of the liquid, g is the acceleration due to gravity, h is the height of the liquid column, and r is the

radius of the capillary tube.

2. **Maximum bubble pressure method:** This method involves measuring the maximum pressure required to blow a bubble from the end of a capillary tube immersed in the liquid. The surface tension is then calculated using the formula: **Surface tension** = $(4P_m)/d$, where P_m is the maximum pressure and d is the diameter of the capillary tube.
3. **Drop weight method:** This method involves measuring the weight of a droplet of liquid that detaches from the end of a capillary tube. The surface tension is then calculated using the formula: **Surface tension** = mg/d , where m is the mass of the droplet, g is the acceleration due to gravity, and d is the diameter of the capillary tube.
4. **Wilhelmy plate method:** This method involves immersing a thin, flat plate into the liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula: **Surface tension** = F/L , where F is the force and L is the perimeter of the plate.
5. **Du Nouy ring method:** This method involves immersing a ring into the liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula: **Surface tension** = $(F - F_c)/4R$, where F is the force, F_c is the correction factor, and R is the radius of the ring.

Each of these methods has its own advantages and limitations, and the choice of method depends on the specific requirements of the experiment. It is important to carefully calibrate the instruments and follow the experimental procedure to obtain accurate results.

6. Determination of Viscosity of a given liquid by viscometer

Viscosity is a measure of a fluid's resistance to flow. It is commonly measured using a viscometer, which is an instrument that measures the force required to move a fluid through a tube or capillary. Here are the steps to determine the viscosity of a given liquid using a viscometer:

1. Fill the viscometer with the liquid to be tested.
2. Place the viscometer in a constant temperature bath to ensure that the temperature of the liquid remains constant throughout the experiment.
3. Measure the time it takes for the liquid to flow through the viscometer tube or capillary.
4. Repeat the measurement several times to obtain an average value.
5. Use the viscometer calibration data and the measured flow time to calculate the viscosity of the liquid.

It is important to note that the viscosity of a liquid can vary with temperature, so it is important to control the temperature during the experiment. Additionally, the viscometer should be properly calibrated to ensure accurate results.

Determination of the strength of Ferrous ammonium sulfate using external indicator

1. Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate.
2. Its chemical formula is $(\text{NH}_4)_2\text{Fe}(\text{SO}_4) \cdot 2.6\text{H}_2\text{O}$.
3. The strength of ferrous ammonium sulfate can be determined using an external indicator.
4. An external indicator is a substance that changes color when the reaction is complete.
5. One common external indicator used in this experiment is potassium permanganate (KMnO_4).
6. Potassium permanganate is a strong oxidizing agent that reacts with ferrous ions to produce ferric ions and manganese dioxide.
7. The reaction is complete when the pink color of the potassium permanganate solution disappears.
8. The strength of the ferrous ammonium sulfate solution can be calculated by measuring the volume of potassium permanganate solution required to complete the reaction.
9. This method is known as a titration.
10. The strength of the ferrous ammonium sulfate solution is expressed in terms of its molarity, which is the number of moles of solute per liter of solution.

Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is a primary standard substance that can be used to determine the strength of a solution.
2. An internal indicator is used to determine the end point of the titration.
3. The internal indicator used in this experiment is potassium iodide.
4. The reaction between potassium dichromate and potassium iodide produces iodine, which can be detected by its characteristic color.
5. The end point of the titration is reached when all the iodine has been consumed and the solution turns from a yellow color to a colorless solution.
6. The strength of the potassium dichromate solution can be calculated using the volume of the solution used and the concentration of the iodine solution.

Determination of Available Chlorine in Bleaching Powder

1. The available chlorine present in a bleaching powder sample is determined iodometrically by treating its solution with an excess of potassium iodide solution in an acidic medium.
2. The liberated iodine (I_2) is treated with sodium thiosulphate ($Na_2S_2O_3$) solution using freshly prepared starch solution as an indicator to be added near the endpoint.
3. The chlorine present in the bleaching powder gets reduced with time.
4. The chlorine that is liberated is called available chlorine, and the calculation of available chlorine is called iodometry.
5. Available chlorine is determined by adding a measured amount of sample solution to an acidified solution of potassium iodide, liberating an equivalent amount of iodine. This free iodine is determined by titration with standardized sodium thiosulfate solution, using a starch indicator endpoint.

Determination of Chloride Content in Water Sample

1. Chloride is one of the major anions found in water and wastewater.
2. The chloride content of water can be determined by titration with silver nitrate solution, using potassium chromate as an indicator.
3. The reaction between silver nitrate and chloride ions produces a white precipitate of silver chloride.
4. The end point of the titration is indicated by the formation of a red-brown color, due to the reaction between excess silver ions and chromate ions.
5. The concentration of chloride ions in the water sample can be calculated from the volume of silver nitrate solution used in the titration.
6. The presence of other halide ions, such as bromide and iodide, can interfere with the determination of chloride content.
7. To overcome this interference, the water sample can be treated with nitric acid and potassium permanganate to oxidize the interfering halide ions.
8. The determination of chloride content in water is important for monitoring the quality of drinking water and wastewater.
9. High levels of chloride in water can indicate contamination from sources such as industrial discharges or road salt.
10. The maximum allowable concentration of chloride in drinking water is 250 mg/L, according to the World Health Organization.

Preparation of Phenol formaldehyde (PF) resin

Phenol formaldehyde (PF) resin, also known as phenolic resin, is a synthetic polymer obtained by the reaction of phenol or substituted phenol with formaldehyde. PF resins are formed by a step-growth polymerization reaction that can be either acid- or base-catalyzed. The concentration of the reactive form of formaldehyde depends on temperature and pH, as formaldehyde exists predominantly in solution as a dynamic equilibrium of methylene glycol oligomers.

To prepare PF resin, phenol, formaldehyde, water, and a catalyst are mixed in the desired amount, depending on the resin to be formed, and are then heated. The first part of the reaction, at around 70 °C, forms a thick reddish-brown tacky material, which is rich in hydroxymethyl and benzylic ether groups.

PF resins were the first completely synthetic polymers to be commercialized and were used as the basis for Bakelite .

Preparation of Urea Formaldehyde (UF) Resin

Urea Formaldehyde (UF) resin is a type of thermosetting adhesive that is commonly used in the wood industry for bonding particleboard, plywood, and medium-density fiberboard (MDF). It is made by reacting urea and formaldehyde in the presence of a base catalyst, typically ammonia or an amine.

Here are the steps involved in the preparation of UF resin:

1. **Mixing of reactants:** Urea and formaldehyde are mixed in a reactor vessel in a molar ratio of 1:1.5 to 1:2.5, depending on the desired properties of the final resin. A base catalyst, such as ammonia or an amine, is added to the mixture to initiate the reaction.
2. **Condensation reaction:** The mixture is heated to a temperature of 70-90°C and stirred for several hours to allow the condensation reaction to take place. During this reaction, the urea and formaldehyde molecules combine to form larger molecules, releasing water as a byproduct.
3. **Adjustment of pH:** The pH of the reaction mixture is adjusted to a value of 7-8 by adding an acid, such as hydrochloric acid or sulfuric acid. This neutralizes the base catalyst and stops the condensation reaction.
4. **Concentration:** The reaction mixture is concentrated by evaporating the water until the desired solids content is reached. This can be done by heating the mixture under reduced pressure or by passing it through a spray dryer.
5. **Cooling and storage:** The resulting UF resin is cooled and stored in airtight containers until it is ready to be used. The resin has a shelf life of several months if stored properly.

These are the basic steps involved in the preparation of UF resin. The exact conditions and proportions of reactants may vary depending on the desired properties of the final resin. It is important to carefully control the reaction conditions to ensure that the resin has the desired properties and performance.

Preparation of Adipic acid

Adipic acid is an organic compound with the formula $(\text{CH}_2)_4(\text{COOH})_2$. It is the most important dicarboxylic acid from an industrial perspective, with about 2.5 billion kilograms of this white crystalline powder being produced annually, mainly as a precursor for the production of nylon .

Adipic acid is produced from a mixture of cyclohexanone and cyclohexanol called KA oil, the abbreviation of ketone-alcohol oil. The KA oil is oxidized with nitric acid to give adipic acid, via a multistep pathway .

Preparation of Paracetamol

Paracetamol is prepared from p-aminophenol by acetylating it with acetic anhydride in the presence of 3-4 drops of concentrated sulphuric acid as a catalyst .

Paracetamol is available as 325-mg or 500-mg tablets, which may include calcium stearate or magnesium stearate, cellulose, docusate sodium and sodium benzoate or sodium lauryl sulfate, starch, hydroxypropyl methylcellulose, propylene glycol, sodium starch glycolate, polyethylene glycol and Red #40 .

14. Determination of Cell Conductance of a solution

Cell conductance is a measure of a solution's ability to conduct electricity. It is determined by the concentration of ions in the solution and their mobility. Here are the steps to determine the cell conductance of a solution:

1. Prepare the solution whose conductance is to be determined.
2. Clean the conductivity cell with distilled water and dry it.
3. Fill the cell with the solution.
4. Connect the cell to a conductivity meter.
5. Switch on the meter and allow it to stabilize.
6. Record the cell conductance as displayed on the meter.

It is important to calibrate the conductivity meter with a standard solution before use to ensure accurate results. The cell constant of the conductivity cell should also be determined to convert the conductance reading to conductivity. The cell constant is determined by measuring the conductance of a solution of

known conductivity and dividing the conductivity by the conductance reading. The cell constant is then used to convert the conductance reading of the unknown solution to conductivity by multiplying the conductance reading by the cell constant.

Determination of Rate constant of hydrolysis of esters

1. The rate constant of hydrolysis of esters can be determined by measuring the rate of reaction at different concentrations of the reactants.
2. The reaction between an ester and water to produce a carboxylic acid and an alcohol is known as hydrolysis.
3. The rate of hydrolysis of an ester can be determined by measuring the change in concentration of one of the reactants or products over time.
4. The rate law for the hydrolysis of an ester can be written as: $\text{rate} = k[\text{ester}]$, where k is the rate constant.
5. The rate constant, k , can be determined by measuring the initial rate of the reaction at different initial concentrations of the ester and water.
6. A graph of the initial rate versus the initial concentration of the ester can be plotted, and the slope of the line will give the value of the rate constant, k .
7. The units of the rate constant will depend on the order of the reaction with respect to the ester and water.
8. The order of the reaction can be determined by varying the concentration of one of the reactants while keeping the concentration of the other reactant constant, and observing the effect on the initial rate of the reaction.
9. The hydrolysis of esters is typically a first-order reaction with respect to the ester and zero-order with respect to water.
10. The rate constant of the hydrolysis of an ester can also be determined by monitoring the change in pH of the reaction mixture over time, as the carboxylic acid produced in the reaction will affect the pH of the solution.
11. The rate constant can also be determined by measuring the change in conductivity of the reaction mixture over time, as the ions produced in the reaction will affect the conductivity of the solution.
12. The hydrolysis of esters is an important reaction in organic chemistry, and the determination of the rate constant can provide valuable information about the reactivity of different esters.
13. The rate constant of the hydrolysis of an ester can be affected by factors such as temperature, solvent, and the presence of catalysts.
14. The Arrhenius equation can be used to determine the activation energy of the hydrolysis reaction by measuring the rate constant at different temperatures.
15. The determination of the rate constant of the hydrolysis of esters is an important tool in the study of reaction kinetics and the factors that affect

the rate of chemical reactions.

Element detection and identification of functional groups in organic compounds

1. **Element detection** refers to the process of identifying the presence of specific elements in a given compound.
2. **Functional groups** are specific groups of atoms within a molecule that are responsible for the characteristic chemical reactions of that molecule.
3. There are several methods for detecting elements and identifying functional groups in organic compounds, including:
 - **Combustion analysis:** This method involves burning a sample of the compound and analyzing the resulting products to determine the presence of elements such as carbon, hydrogen, and nitrogen.
 - **Infrared spectroscopy:** This technique uses infrared radiation to identify the presence of specific functional groups based on their characteristic absorption patterns.
 - **Mass spectrometry:** This method involves ionizing a sample of the compound and analyzing the resulting fragments to determine the presence of specific elements and functional groups.
 - **Nuclear magnetic resonance (NMR) spectroscopy:** This technique uses the interaction of nuclear spins with a magnetic field to identify the presence of specific functional groups and the connectivity of atoms within a molecule.
4. These methods can be used individually or in combination to provide a comprehensive analysis of the elements and functional groups present in an organic compound.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose any 10 experiments from the list provided.
- The instructor also has the option to change any two of the experiments from the list.
- This allows the instructor to tailor the experiments to the needs and interests of the students.
- It is important for students to carefully read and understand the instructions for each experiment.
- Students should also be prepared for the possibility that the instructor may change some of the experiments.

Course Outcomes

Course outcomes are statements that describe the knowledge, skills, and abilities that students are expected to demonstrate upon completion of a course. These outcomes serve as a guide for instructors in designing course content and assessments, and for students in understanding the goals of the course.

Some key points to consider when developing course outcomes include:

1. Outcomes should be specific, measurable, and achievable within the scope of the course.
2. Outcomes should align with the overall goals of the program or curriculum.
3. Outcomes should be written in clear, concise language that is easily understood by students.
4. Outcomes should be reviewed and updated regularly to ensure they remain relevant and accurate.

Effective course outcomes can help to improve student learning and success by providing clear expectations and goals for the course. They can also serve as a valuable tool for instructors in evaluating the effectiveness of their teaching and course design.

Upon completion of the course, the student should be able to:

1. Demonstrate a comprehensive understanding of the course material.
2. Apply the knowledge and skills learned in the course to real-world situations.
3. Analyze and evaluate information related to the course material.
4. Communicate effectively, both orally and in writing, about the course material.
5. Work collaboratively with others to complete course-related tasks.
6. Demonstrate critical thinking and problem-solving skills.
7. Use technology effectively to enhance learning and complete course-related tasks.
8. Demonstrate ethical behavior and professionalism in all course-related activities.
9. Engage in continuous learning and professional development related to the course material.

Course Outcomes Bloom's

Course outcomes are statements that describe the knowledge, skills, and abilities that students should have acquired by the end of a course. These outcomes are typically aligned with the course objectives and are used to assess the effectiveness of the course in achieving its goals.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. It is often used to design

course outcomes and assessments. The taxonomy consists of six levels, arranged in a hierarchy from lower-order thinking skills to higher-order thinking skills:

1. **Remembering:** The ability to recall or recognize previously learned information.
2. **Understanding:** The ability to comprehend the meaning of material.
3. **Applying:** The ability to use learned material in new and concrete situations.
4. **Analyzing:** The ability to break down material into its component parts and understand its structure.
5. **Evaluating:** The ability to make judgments about the value of material for a given purpose.
6. **Creating:** The ability to put parts together to form a new whole.

Course outcomes can be written to target specific levels of Bloom's Taxonomy. For example, an outcome that targets the "applying" level might read: "By the end of this course, students will be able to apply the principles of supply and demand to analyze market behavior." This outcome requires students to use their understanding of supply and demand to solve problems in a new context.

In summary, course outcomes are statements that describe what students should be able to do by the end of a course, and Bloom's Taxonomy is a framework that can be used to design and assess these outcomes. By aligning course outcomes with the levels of Bloom's Taxonomy, instructors can ensure that their courses are challenging and engaging for students, and that assessments accurately measure student learning.

Level

- A level is a tool used to determine if a surface is horizontal or vertical.
- It consists of a small glass tube filled with liquid, usually alcohol or a colored spirit, with an air bubble inside.
- When the level is placed on a surface, the bubble will move to the highest point of the tube, indicating if the surface is level or not.
- There are different types of levels, including spirit levels, laser levels, and digital levels.
- Levels are commonly used in construction, carpentry, and surveying, among other fields.
- It is important to use a level when installing or building structures to ensure they are straight and stable.
- Levels can also be used to measure angles and slopes.
- Some levels come with additional features, such as rulers or magnets, to make them more versatile and useful.

CO-1: Understanding the use of different analytical instruments

Analytical instruments are used to measure physical, chemical, and biological properties of substances. These instruments are used in a variety of fields, including chemistry, biology, physics, and engineering. Some common analytical instruments include:

1. **Spectrophotometers:** These instruments measure the absorption or transmission of light by a sample. They are commonly used to determine the concentration of a substance in a solution.
2. **Chromatographs:** These instruments separate mixtures into their individual components. They are commonly used to identify and quantify the components of a mixture.
3. **Mass spectrometers:** These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify the chemical composition of a sample.
4. **Electrochemical instruments:** These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.
5. **Thermal analysis instruments:** These instruments measure the thermal properties of a sample. They are commonly used to determine the melting point, boiling point, and heat capacity of a substance.

Understanding the use of these instruments is important for conducting accurate and reliable experiments. By selecting the appropriate instrument for a given experiment, researchers can obtain accurate and precise measurements of the properties of a substance. This knowledge is essential for making informed decisions and drawing accurate conclusions from experimental data.

CO2: Measuring Molecular/System Properties such as Surface Tension

Carbon dioxide (CO₂) is a colorless, odorless gas that is naturally present in the Earth's atmosphere. It is an important greenhouse gas that helps regulate the temperature of the planet. However, human activities such as burning fossil fuels and deforestation have increased the concentration of CO₂ in the atmosphere, contributing to global warming.

Surface tension is a property of liquids that arises from the cohesive forces between molecules at the surface of a liquid. It is the force that allows insects to walk on water and causes water droplets to form beads on a surface.

To measure the surface tension of CO₂, the following steps can be taken:

1. Prepare a container with a flat surface and fill it with liquid CO₂.
2. Place a small, flat object (such as a coin) on the surface of the liquid CO₂.
3. Observe the behavior of the object. If it floats, the surface tension of the liquid CO₂ is strong enough to support the weight of the object.
4. To determine the exact value of the surface tension, a more precise method such as the maximum bubble pressure method or the drop weight method can be used.

It is important to note that the surface tension of CO₂ can vary depending on factors such as temperature and pressure. Therefore, it is important to control these variables when measuring the surface tension of CO₂.

Viscosity, Conductance of Solution, Chloride and Iron Content in Water

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.
- The viscosity of formation water is a function of pressure, temperature, and dissolved solids.
- In general, brine viscosity increases with increasing pressure, increasing salinity, and decreasing temperature.
- Dissolved gas in the formation water at reservoir conditions generally results in a negligible effect on water viscosity.

Conductance of Solution

- Conductivity is a measure of a solution's ability to conduct an electric current.
- High quality deionized water has a conductivity of about 0.05 S/cm at 25 °C, typical drinking water is in the range of 200–800 S/cm, while sea water is about 50 mS/cm (or 0.05 S/cm).
- The more chloride ions present in the water, the higher the conductivity.
- In general, the conductivity of waterways in the United States varies from 50 to 1500 µmhos/cm, and inland freshwater lake studies reveal a conductivity of 150 to 500 µmhos/cm.

Chloride and Iron Content in Water

- Chloride ions are approximately 1.9 percent of the mass of seawater.
- The presence of chloride ions in water can affect its conductivity, as mentioned above.
- Iron content in water can affect its color, taste, and odor, and can also stain laundry and fixtures.
- Iron can be present in water in two forms: soluble ferrous iron or insoluble ferric iron.

- Water treatment methods for removing iron from water include ion exchange, oxidation and filtration, and sequestration.

K3

- K3 is a term that can refer to several different things, depending on the context in which it is used.
- It can refer to a mountain in the Karakoram range, also known as Broad Peak.
- It can also refer to a visa category for spouses of U.S. citizens.
- In mathematics, K3 can refer to a K3 surface, which is a complex algebraic surface with certain properties.
- In music, K3 is a Belgian-Dutch girl group.
- In transportation, K3 can refer to a class of passenger carriages used by the Indian Railways.
- In summary, K3 can have various meanings and it is important to understand the context in which it is used to determine its specific meaning.

CO-3 Measure the hardness and alkalinity of the water. K3

Hardness and alkalinity are two important parameters used to measure the quality of water. Here are some key points to remember when measuring the hardness and alkalinity of water:

1. **Hardness** refers to the concentration of calcium and magnesium ions in water. These ions can cause scaling in pipes and appliances, and can also affect the taste of the water.
2. **Alkalinity** refers to the ability of water to neutralize acids. It is a measure of the buffering capacity of water, which is important for maintaining a stable pH.
3. To measure the hardness of water, a titration method can be used. This involves adding a chemical reagent to a water sample until a color change occurs, indicating the presence of calcium and magnesium ions.
4. To measure the alkalinity of water, a similar titration method can be used. In this case, a chemical reagent is added to the water sample until a pH change occurs, indicating the presence of buffering compounds.
5. It is important to measure both the hardness and alkalinity of water, as they can affect the overall quality of the water and its suitability for various uses.
6. Regular monitoring of water hardness and alkalinity can help to identify changes in water quality and take appropriate action to address any issues.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an organic compound that is also known as carbolic acid. It is a weak acid that mostly forms phenoxide ions by dropping one positive hydrogen ion (H^+) from the hydroxyl group. There are several methods for the preparation of phenol, including:

1. **From Haloarenes:** Chlorobenzene is an example of a haloarene which is formed by the monosubstitution of the benzene ring. When chlorobenzene is fused with sodium hydroxide at 623K and 320 atm, sodium phenoxide is produced .
2. **From Benzene Sulphonic Acid:** At 573 – 623K, sodium phenoxide is produced by fusing the sodium salt of benzene sulphonic acid with sodium hydroxide. Sodium phenoxide is converted to phenol when it is treated with dilute acid or carbon dioxide.
3. **From Coal Tar:** Phenol is extracted from the middle oil fraction for commercial purposes at 443 – 503K.
4. **Raschig's Process:** The reaction of benzene, hydrogen chloride and air at 523K produces phenol.
5. **From Cumene:** Phenol can be prepared in large quantities by the air oxidation of cumene.
6. **From Diazonium Salts:** Small amounts of phenol can be prepared by the hydrolysis of diazonium salts.
7. **From Phenylboronic Acid:** Small amounts of phenol can be prepared by the peroxide oxidation of phenylboronic acid.

Formaldehyde & Urea Formaldehyde Resin

- Formaldehyde is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.
- Urea-formaldehyde (UF), also known as urea-methanal, is a nontransparent thermosetting resin or polymer. It is produced from urea and formaldehyde.
- These resins are used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- Using formaldehyde and urea as raw materials, a stable urea-formaldehyde resin (UF) is synthesized by the “alkali-acid-alkali” method .
- Unlike most thermosetting resins, UF often shows the appearance of crystal domains .

Adipic Acid

- Adipic acid dihydrazide can be used as a modifier to reduce free formaldehyde in urea-formaldehyde (UF) resin.

Paracetamol

K3

- K3 is a synthetic, artificially produced form of vitamin K that doesn't occur naturally .
- This is unlike the other two forms of vitamin K — vitamin K1, known as phyloquinone, and vitamin K2, called menaquinone .
- Vitamin K3 may be converted into K2 in your liver .
- Vitamin K3 is also known as menadione .
- Vitamin K is important for blood clotting and bone health .
- Although vitamin K3 has demonstrated anticancer and antibacterial properties in test-tube studies, it has potential side effects .
- K3 is also the name of a form used by the IRS, Schedule K-3 (Form 1065) 2022 Partner's Share of Income, Deductions, Credits, etc.— International Department of the Treasury Internal Revenue Service .

CO-5 Estimate the rate constant of reaction. K3

The rate constant of a reaction, denoted by the symbol k , is a measure of the speed of a chemical reaction. It is defined as the proportionality constant between the rate of the reaction and the concentrations of the reactants raised to their respective powers, as given by the rate law of the reaction.

To estimate the rate constant of a reaction, the following steps can be followed:

1. Determine the order of the reaction with respect to each reactant by performing experiments in which the concentration of one reactant is varied while the concentrations of the other reactants are kept constant.
2. Write the rate law for the reaction using the determined orders of reaction.
3. Measure the initial rate of the reaction for several sets of initial concentrations of the reactants.
4. Substitute the initial concentrations and the initial rate of the reaction into the rate law to obtain a set of simultaneous equations.
5. Solve the simultaneous equations to obtain the value of the rate constant, k .

It is important to note that the value of the rate constant, k , is specific to a particular reaction and is dependent on factors such as temperature and the

presence of a catalyst. The units of the rate constant also depend on the order of the reaction.

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ENGINEERING PHYSICS LAB

Engineering Physics Lab is a course that is designed to provide students with hands-on experience in the application of physics principles in engineering. The course typically includes experiments that demonstrate the principles of mechanics, optics, electricity, and magnetism.

Some of the key objectives of the Engineering Physics Lab course are: - To provide students with practical experience in the application of physics principles to engineering problems. - To develop students' skills in experimental design, data analysis, and scientific communication. - To reinforce the theoretical concepts covered in the lecture component of the course.

The experiments conducted in the Engineering Physics Lab may vary depending on the institution and the specific course curriculum. Some common experiments that may be included in the course are: - Measurement of physical quantities such as length, mass, and time using various instruments. - Study of the properties of waves, including reflection, refraction, and interference. - Investigation of the behavior of electric circuits, including Ohm's law and Kirchhoff's laws. - Study of the properties of magnetic fields and their interaction with electric currents.

The Engineering Physics Lab course typically includes a combination of individual and group work, with students working in teams to design and conduct experiments, analyze data, and present their results. The course may also include written reports and oral presentations as part of the assessment.

Overall, the Engineering Physics Lab course is an important component of an engineering education, providing students with the opportunity to apply their theoretical knowledge to practical problems and develop key skills in experimental design and data analysis.

Course Objectives:

1. To provide a comprehensive understanding of the subject matter.

2. To develop critical thinking and analytical skills.
3. To enhance the ability to apply theoretical concepts to practical situations.
4. To encourage independent learning and research.
5. To prepare students for further academic or professional pursuits in the field.
6. To foster an appreciation for the subject and its relevance to various aspects of life.
7. To promote effective communication and collaboration skills.
8. To instill ethical and responsible behavior in the pursuit of knowledge.

1. To enable the students to understand about the fundamental concepts of analytical instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are commonly used in scientific research, quality control, and medical diagnosis. Some of the fundamental concepts of analytical instruments include:

- **Accuracy:** This refers to how close the measurement is to the true value of the substance being analyzed.
- **Precision:** This refers to how consistent the measurements are when the same substance is analyzed multiple times.
- **Sensitivity:** This refers to the ability of the instrument to detect small changes in the substance being analyzed.
- **Selectivity:** This refers to the ability of the instrument to distinguish between different substances in a mixture.
- **Calibration:** This is the process of adjusting the instrument to ensure that it is providing accurate and precise measurements.
- **Maintenance:** This refers to the regular upkeep and repair of the instrument to ensure that it is functioning properly.

Understanding these fundamental concepts is essential for students to effectively use and interpret the results from analytical instruments.

2. Analysis of Chloride Content, Hardness, and Alkalinity of Water

- **Chloride Content:** Chloride is one of the major anions found in water and wastewater. The chloride test measures the level of chloride ions in water. High levels of chloride can indicate pollution from human or animal waste, industrial processes, or saltwater intrusion.
- **Hardness:** Water hardness is a measure of the amount of calcium and magnesium ions in water. These ions can cause problems in water systems by forming scale deposits in pipes and boilers. Hardness is usually expressed in terms of calcium carbonate (CaCO_3) equivalents.

- **Alkalinity:** Alkalinity is a measure of the water's ability to neutralize acids. It is important in maintaining the pH balance of water and preventing corrosion in water systems. Alkalinity is usually expressed in terms of milligrams per liter (mg/L) of calcium carbonate (CaCO_3) equivalents.

These parameters are important in the analysis of water quality and can provide valuable information about the sources of pollution and the suitability of water for various uses. It is important for students to understand these concepts in order to properly analyze and interpret water quality data.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of a liquid.

- **pH:** pH is a measure of the acidity or basicity of a solution. It is defined as the negative logarithm of the hydrogen ion concentration. The pH scale ranges from 0 to 14, with 7 being neutral. A pH less than 7 is acidic, while a pH greater than 7 is basic. pH can be measured using a pH meter or pH paper.
- **Surface tension:** Surface tension is the property of a liquid that allows it to resist an external force. It is caused by the cohesive forces between the molecules of the liquid. Surface tension can be measured using a number of methods, including the drop weight method, the maximum bubble pressure method, and the Wilhelmy plate method.
- **Viscosity:** Viscosity is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate. Viscosity can be measured using a number of methods, including the capillary tube method, the falling ball method, and the rotational viscometer method. Viscosity is affected by temperature, with most fluids becoming less viscous as the temperature increases.

These properties of liquids are important in many fields, including chemistry, physics, and engineering. Understanding how to measure and interpret these properties can help students better understand the behavior of liquids in various situations.

4. To enable the students to understand about the preparation of different resins.

Resins are solid or highly viscous substances that are typically convertible into polymers. They are usually mixtures of organic compounds, mainly terpenes and their derivatives. Here are some points to consider when preparing different resins:

1. **Types of resins:** There are several types of resins, including natural resins, synthetic resins, and modified natural resins. Each type of resin has its own unique properties and preparation methods.

2. **Raw materials:** The raw materials used in the preparation of resins vary depending on the type of resin being prepared. For example, natural resins are typically derived from plant sources, while synthetic resins are made from chemicals.
3. **Preparation methods:** There are several methods for preparing resins, including solvent extraction, precipitation, and polymerization. The method used will depend on the type of resin being prepared and the desired properties of the final product.
4. **Safety precautions:** The preparation of resins can involve the use of hazardous chemicals and high temperatures. It is important to follow proper safety precautions, such as wearing protective equipment and working in a well-ventilated area.
5. **Quality control:** Quality control is an important aspect of resin preparation. This can involve testing the properties of the final product to ensure that it meets the desired specifications.

In summary, the preparation of different resins involves understanding the types of resins, the raw materials used, the preparation methods, safety precautions, and quality control measures. By following these guidelines, students can learn how to prepare different resins effectively.

5. To enable the students to understand about the synthesis of organic compounds such as adipic acid and paracetamol by conventional and green route.

- Adipic acid is an important organic compound that is used in the production of nylon. It can be synthesized through a conventional route, which involves the oxidation of cyclohexanol or cyclohexanone with nitric acid. This process produces large amounts of nitrous oxide, a potent greenhouse gas.
- A green route for the synthesis of adipic acid involves the use of hydrogen peroxide as the oxidizing agent, which produces water as the only byproduct. This process is more environmentally friendly and sustainable.
- Paracetamol, also known as acetaminophen, is a widely used pain reliever and fever reducer. It can be synthesized through a conventional route, which involves the nitration of phenol followed by the reduction of the nitro group to an amine and then acetylation.
- A green route for the synthesis of paracetamol involves the use of enzymes to catalyze the reaction. This process is more environmentally friendly and produces less waste.
- Green chemistry is an approach to chemical synthesis that aims to reduce the environmental impact of chemical processes. It involves the use of

sustainable and renewable resources, the reduction of waste and the use of safer and less toxic chemicals.

- The use of green routes for the synthesis of organic compounds such as adipic acid and paracetamol is an important step towards a more sustainable and environmentally friendly chemical industry.

LIST OF EXPERIMENTS

1. **The Milgram Experiment:** A series of social psychology experiments conducted by Yale University psychologist Stanley Milgram in the 1960s, which measured the willingness of study participants to obey an authority figure who instructed them to perform acts conflicting with their personal conscience.
2. **The Stanford Prison Experiment:** A social psychology experiment conducted at Stanford University in 1971, in which college students were randomly assigned to play the roles of prisoners or guards in a simulated prison environment. The study aimed to investigate the psychological effects of perceived power.
3. **The Asch Conformity Experiment:** A series of studies conducted by Solomon Asch in the 1950s, which investigated the extent to which social pressure from a majority group could affect a person to conform.
4. **The Pavlov's Dog Experiment:** A series of experiments conducted by Russian physiologist Ivan Pavlov in the 1890s, in which he demonstrated that dogs could be trained to associate a neutral stimulus, such as the sound of a bell, with an automatic behavior, such as salivating when presented with food.
5. **The Marshmallow Test:** A study on delayed gratification conducted by psychologist Walter Mischel in the 1960s and 1970s, in which children were offered a choice between a small immediate reward or a larger reward if they waited for a short period of time.

1. Calibration of Analytical Equipment and Apparatus

Calibration is the process of verifying the accuracy of a measuring instrument by comparing it with a reference standard. This is important for ensuring that the results obtained from the instrument are accurate and reliable.

Here are some key points to consider when calibrating analytical equipment and apparatus:

- Calibration should be performed regularly to ensure that the instrument is functioning properly and providing accurate results.
- The frequency of calibration will depend on factors such as the type of instrument, its usage, and the environment in which it is used.
- Calibration should be performed by trained personnel using appropriate reference standards and procedures.

- Records of calibration should be maintained to provide evidence of the instrument's accuracy and reliability.
- If an instrument is found to be out of calibration, it should be adjusted or repaired before being used again.

In summary, calibration is an essential step in ensuring the accuracy and reliability of analytical equipment and apparatus. It should be performed regularly by trained personnel using appropriate reference standards and procedures. Records of calibration should be maintained to provide evidence of the instrument's accuracy and reliability. If an instrument is found to be out of calibration, it should be adjusted or repaired before being used again.

2. Determination of Hardness of water sample by EDTA method

- Hardness of water is determined by the presence of calcium and magnesium ions in the water.
- EDTA (ethylenediaminetetraacetic acid) is a chelating agent that can bind to these ions and form a complex.
- In the EDTA method, a solution of EDTA is added to a water sample until all the calcium and magnesium ions have formed complexes with the EDTA.
- The amount of EDTA used is then used to calculate the hardness of the water sample.
- The endpoint of the titration is determined using an indicator such as Eriochrome Black T, which changes color when all the calcium and magnesium ions have been complexed.
- This method is a simple and accurate way to determine the hardness of a water sample.

3. Determination of Alkalinity of water sample

Alkalinity is a measure of the ability of water to neutralize acids. It is an important parameter in water quality analysis, as it provides information about the buffering capacity of water. The following points explain the determination of alkalinity in a water sample:

1. The water sample is collected in a clean container and its pH is measured.
2. A known volume of the water sample is taken and titrated against a standard acid solution, usually sulfuric acid or hydrochloric acid.
3. The titration is carried out using a suitable indicator, such as phenolphthalein or methyl orange, which changes color at the endpoint of the titration.
4. The volume of the acid solution required to reach the endpoint is noted and used to calculate the alkalinity of the water sample.
5. The alkalinity is usually expressed in terms of milligrams of calcium carbonate (CaCO_3) per liter of water.

This is a brief overview of the determination of alkalinity in a water sample.

It is important to follow standard procedures and use accurate equipment to obtain reliable results.

4. Determination of pH by titrimetric method

Titrimetric method is a common laboratory technique used to determine the pH of a solution. The process involves the following steps:

1. A solution of known concentration, called the titrant, is prepared. This is usually a strong acid or base.
2. The solution to be tested, called the analyte, is measured and placed in a container.
3. A few drops of an indicator are added to the analyte. The indicator is a substance that changes color depending on the pH of the solution.
4. The titrant is slowly added to the analyte while stirring. The indicator changes color when the pH of the solution reaches a certain value.
5. The volume of titrant added is recorded when the indicator changes color. This is called the endpoint.
6. The pH of the solution can be calculated using the volume of titrant added and the concentration of the titrant.

This method is accurate and reliable, but it requires careful preparation and execution. It is commonly used in laboratories and in industrial processes to determine the pH of solutions.

5. Determination of surface tension of given liquid

Surface tension is a property of liquids that arises due to the cohesive forces between the liquid molecules. It is defined as the force acting per unit length on the surface of the liquid. There are several methods to determine the surface tension of a given liquid, including:

1. **Capillary rise method:** This method involves measuring the height to which a liquid rises in a capillary tube due to the surface tension of the liquid. The surface tension is then calculated using the formula: **Surface tension** = $(\rho g h r)/2$, where ρ is the density of the liquid, g is the acceleration due to gravity, h is the height of the liquid column, and r is the radius of the capillary tube.
2. **Maximum bubble pressure method:** This method involves measuring the maximum pressure required to blow a bubble from a submerged orifice. The surface tension is then calculated using the formula: **Surface tension** = $(4P_m)/d$, where P_m is the maximum pressure and d is the diameter of the orifice.
3. **Drop weight method:** This method involves measuring the weight of drops of liquid that detach from the end of a thin tube. The surface tension is then calculated using the formula: **Surface tension** = mg/d ,

where m is the mass of the drop, g is the acceleration due to gravity, and d is the diameter of the tube.

4. **Du Nouy ring method:** This method involves measuring the force required to detach a ring from the surface of a liquid. The surface tension is then calculated using the formula: $\text{Surface tension} = F/4r$, where F is the force and r is the radius of the ring.
5. **Wilhelmy plate method:** This method involves measuring the force required to detach a flat plate from the surface of a liquid. The surface tension is then calculated using the formula: $\text{Surface tension} = F/2l$, where F is the force and l is the length of the plate.

These are some of the common methods used to determine the surface tension of a given liquid. Each method has its own advantages and limitations, and the choice of method depends on the specific requirements of the experiment.

6. Determination of Viscosity of a given liquid by viscometer

Viscosity is a measure of a fluid's resistance to flow. It is commonly measured using a device called a viscometer. Here are the steps to determine the viscosity of a given liquid using a viscometer:

1. **Select the appropriate viscometer:** Different types of viscometers are available for different types of liquids and measurement ranges. Choose the one that is suitable for the liquid you want to measure.
2. **Prepare the sample:** The liquid sample should be free of air bubbles and at a consistent temperature. If necessary, filter the sample to remove any impurities.
3. **Fill the viscometer:** Follow the manufacturer's instructions to fill the viscometer with the liquid sample. Make sure the liquid level is within the specified range.
4. **Measure the flow time:** Start the timer when the liquid begins to flow and stop it when the liquid reaches the specified mark. Record the flow time.
5. **Calculate the viscosity:** Use the manufacturer's instructions to calculate the viscosity of the liquid based on the flow time and other relevant factors such as temperature and density.
6. **Repeat the measurement:** To ensure accuracy, repeat the measurement several times and take the average of the results.

It is important to follow the manufacturer's instructions carefully and to use the appropriate viscometer for the liquid being measured. Proper sample preparation and accurate measurement of the flow time are also crucial for obtaining reliable results.

7. Determination of the strength of Ferrous ammonium sulfate using external indicator

1. Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate.
2. It is commonly used as a standard solution in redox titrations due to its stability and the ease of preparation.
3. The strength of a ferrous ammonium sulfate solution can be determined by titrating it against a standard solution of potassium permanganate (KMnO_4) using an external indicator.
4. The external indicator used in this titration is potassium ferricyanide ($\text{K}_3\text{Fe}(\text{CN})_6$).
5. The reaction between ferrous ammonium sulfate and potassium permanganate is a redox reaction, where potassium permanganate is reduced to manganese sulfate (MnSO_4) and ferrous ammonium sulfate is oxidized to ferric ammonium sulfate.
6. The end point of the titration is indicated by the appearance of a blue color, due to the formation of Prussian blue, when all the ferrous ions have been oxidized and an excess of potassium permanganate is present.
7. The strength of the ferrous ammonium sulfate solution can be calculated using the volume of potassium permanganate solution used and the molar concentration of the potassium permanganate solution.

8. Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is an oxidizing agent that can be used to determine the strength of a substance using an internal indicator.
2. An internal indicator is a substance that changes color to indicate the end point of a titration.
3. In the case of potassium dichromate, the internal indicator used is usually diphenylamine sulfonate.
4. The procedure involves adding a known volume of potassium dichromate solution to the substance being tested, and then adding the internal indicator.
5. The solution is then titrated with a reducing agent until the color change of the internal indicator indicates that the end point has been reached.
6. The volume of reducing agent used can then be used to calculate the strength of the potassium dichromate solution.
7. This method is commonly used in the analysis of iron, copper, and other metals in various samples.
8. It is important to note that the accuracy of this method depends on the purity of the potassium dichromate solution and the correct use of the internal indicator.

9. Determination of available chlorine in bleaching powder

Bleaching powder, also known as calcium hypochlorite, is a chemical compound with the formula $\text{Ca}(\text{ClO})_2$. It is widely used as a disinfectant and bleaching agent. The available chlorine in bleaching powder refers to the amount of chlorine that can be released as a disinfectant when the powder is dissolved in water.

Here are the steps to determine the available chlorine in bleaching powder:

1. Weigh accurately about 1g of the sample of bleaching powder and transfer it to a conical flask.
2. Add 100 ml of distilled water to the flask and shake well to dissolve the powder.
3. Filter the solution through a filter paper and collect the filtrate in a conical flask.
4. Take 20 ml of the filtrate in a conical flask and add 1 ml of potassium iodide solution and 2 ml of acetic acid to it.
5. Titrate the solution against sodium thiosulphate solution using starch as an indicator.
6. Repeat the titration until concordant values are obtained.
7. Calculate the available chlorine in the sample using the formula: % available chlorine = $(V * N * 35.5 * 100) / (W * 10)$, where V is the volume of sodium thiosulphate solution used, N is the normality of sodium thiosulphate solution, W is the weight of the sample taken, and 35.5 is the atomic weight of chlorine.

This is a brief overview of the process of determining the available chlorine in bleaching powder. It is important to follow the steps carefully and accurately to obtain reliable results.

10. Determination of chloride content in water sample

1. Chloride content in water can be determined using a titration method known as the Mohr method.
2. The Mohr method involves titrating the water sample with a silver nitrate solution.
3. A chromate indicator is used to signal the endpoint of the titration.
4. The silver nitrate reacts with the chloride ions in the water sample to form a precipitate of silver chloride.
5. The chromate ions react with the excess silver ions to form a red precipitate of silver chromate, indicating the endpoint of the titration.
6. The volume of silver nitrate solution used in the titration can be used to calculate the concentration of chloride ions in the water sample.
7. The concentration of chloride ions can be converted to the mass of chloride in the water sample by multiplying by the molar mass of chloride.
8. The mass of chloride can be expressed as a percentage of the total mass of the water sample to give the percentage chloride content.

9. The Mohr method is a simple and accurate method for determining the chloride content in water samples.
10. Other methods for determining chloride content in water samples include ion chromatography and ion-selective electrode analysis.

11. Preparation of Phenol formaldehyde (PF) resin

Phenol formaldehyde (PF) resin, also known as phenolic resin, is a synthetic polymer obtained by the reaction of phenol or substituted phenol with formaldehyde. PFs were the first commercial synthetic resins (plastics) and were used as the basis for Bakelite.

The preparation of Phenol-formaldehyde resins involves a step-growth polymerization reaction that can be either acid- or base-catalyzed. The concentration of the reactive form of formaldehyde depends on temperature and pH, as formaldehyde exists predominantly in solution as a dynamic equilibrium of methylene glycol oligomers.

To form the resin, phenol, formaldehyde, water, and a catalyst are mixed in the desired amount and then heated. The first part of the reaction, at around 70 °C, forms a thick reddish-brown tacky material, which is rich in hydroxymethyl and benzylic ether groups.

12. Preparation of Urea formaldehyde (UF) resin

Urea formaldehyde (UF) resin is a type of thermosetting polymer that is commonly used in the production of adhesives, finishes, and molded objects. Here are the steps involved in the preparation of UF resin:

1. **Reacting urea and formaldehyde:** The first step in the preparation of UF resin is the reaction of urea and formaldehyde in an aqueous solution. The molar ratio of urea to formaldehyde is typically between 1:1.5 and 1:2.5. The reaction is carried out at a pH of 7-8 and a temperature of 85-95°C.
2. **Condensation:** The next step is the condensation of the reaction mixture. This is done by heating the mixture to 100-105°C and maintaining this temperature for 1-3 hours. During this time, the water content of the mixture is reduced, and the viscosity of the mixture increases.
3. **Cooling and neutralization:** After the condensation step, the mixture is cooled to room temperature. The pH of the mixture is then adjusted to 7-8 by adding an acid or a base.
4. **Filtration:** The mixture is then filtered to remove any impurities or unreacted materials.
5. **Storage:** The resulting UF resin is stored in a cool and dry place until it is ready for use.

These are the basic steps involved in the preparation of UF resin. The exact conditions and parameters may vary depending on the specific application and desired properties of the final product.

13. Preparation of Adipic acid / Paracetamol

Adipic acid and Paracetamol are two different compounds with different preparation methods.

Adipic acid:

Adipic acid is an organic compound with the formula $(\text{CH}_2)_4(\text{COOH})_2$. It is a white crystalline powder and is mainly used in the production of nylon.

1. Adipic acid can be prepared by the oxidation of cyclohexanol or cyclohexanone with nitric acid.
2. Another method involves the oxidation of cyclohexane with air in the presence of a cobalt catalyst.
3. Adipic acid can also be prepared by the hydrolysis of acrylonitrile.

Paracetamol:

Paracetamol, also known as acetaminophen, is an analgesic and antipyretic drug used to relieve pain and reduce fever.

1. Paracetamol can be prepared by the nitration of phenol to produce para-nitrophenol, which is then reduced to para-aminophenol.
2. Para-aminophenol is then acetylated with acetic anhydride to produce paracetamol.
3. Another method involves the direct acetylation of phenol with acetic anhydride in the presence of zinc chloride.

14. Determination of Cell Conductance of a solution

Cell conductance is a measure of a solution's ability to conduct an electric current. It is determined by the concentration and mobility of ions in the solution. Here are the steps to determine the cell conductance of a solution:

1. Prepare the solution whose conductance is to be measured.
2. Clean the conductivity cell with distilled water and dry it.
3. Fill the cell with the solution.
4. Connect the cell to a conductivity meter.
5. Switch on the conductivity meter and allow it to stabilize.
6. Record the conductivity reading displayed on the meter.
7. Repeat the measurement with fresh samples of the solution to obtain an average value.

The cell conductance is calculated by multiplying the conductivity reading by the cell constant, which is determined by calibrating the cell with a solution of

known conductivity. The cell constant is the ratio of the distance between the electrodes to the area of the electrodes. The unit of cell conductance is Siemens (S).

15. Determination of Rate constant of hydrolysis of esters

1. The rate constant of the hydrolysis of esters can be determined by measuring the rate of reaction at different concentrations of the reactants.
2. The rate of reaction can be measured by monitoring the change in concentration of one of the reactants or products over time.
3. The rate law for the hydrolysis of esters can be written as: $\text{rate} = k[\text{ester}]$, where k is the rate constant, $[\text{ester}]$ is the concentration of the ester, and water is the concentration of water.
4. By measuring the rate of reaction at different concentrations of the ester and water, the rate constant k can be determined using the method of initial rates.
5. The method of initial rates involves measuring the initial rate of reaction at different initial concentrations of the reactants.
6. A graph of the initial rate of reaction versus the initial concentration of one of the reactants can be plotted, and the slope of the line can be used to determine the order of the reaction with respect to that reactant.
7. The rate constant k can then be calculated using the rate law and the determined orders of the reaction with respect to the reactants.
8. The hydrolysis of esters is typically a first-order reaction with respect to the ester and water, meaning that the rate of reaction is directly proportional to the concentration of the ester and water.
9. The rate constant k can be used to predict the rate of reaction at different concentrations of the reactants, and can provide information about the mechanism of the reaction.

16. Element detection and identification of functional groups in organic compounds.

1. **Element detection** refers to the process of identifying the presence of specific elements in a given sample of an organic compound.
2. **Functional groups** are specific groups of atoms within a molecule that are responsible for the characteristic chemical reactions of that molecule.
3. The identification of functional groups in organic compounds is important for understanding their chemical properties and reactivity.
4. Common methods for element detection and identification of functional groups in organic compounds include:
 - **Combustion analysis**: used to determine the elemental composition of a sample by burning it in excess oxygen and measuring the resulting products.
 - **Infrared spectroscopy**: used to identify the presence of specific functional groups by measuring the absorption of infrared radiation

- by the sample.
- **Mass spectrometry:** used to determine the molecular weight and elemental composition of a sample by ionizing it and measuring the resulting fragments.
 - **Nuclear magnetic resonance spectroscopy:** used to determine the structure of a molecule by measuring the interaction of its nuclei with a magnetic field.
5. These methods can be used in combination to provide a comprehensive analysis of the elemental composition and functional group identification of an organic compound.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose any 10 experiments from the list provided.
- The instructor also has the option to change any two of the experiments from the list.
- This allows the instructor to tailor the experiments to the needs and interests of the students.
- It is important for students to carefully read and understand the instructions for each experiment.
- Students should also be prepared for the possibility that the instructor may change some of the experiments.

Course Outcomes:

- Define and explain the key concepts and principles of the course.
- Demonstrate an understanding of the course material through application and analysis.
- Communicate effectively in written and oral forms.
- Work collaboratively with others to achieve common goals.
- Develop critical thinking and problem-solving skills.
- Apply ethical principles and values to decision-making.
- Demonstrate an awareness of diverse perspectives and the ability to engage in respectful dialogue.
- Use technology effectively to access, organize, and present information.
- Demonstrate a commitment to lifelong learning and personal and professional development.

Upon completion of the course the student should be able to:

1. Demonstrate a thorough understanding of the course material.
2. Apply the concepts and theories learned in the course to real-world situations.
3. Analyze and evaluate information critically and effectively.

4. Communicate ideas and arguments clearly and effectively in written and oral form.
5. Work collaboratively with others to achieve common goals.
6. Demonstrate ethical and professional behavior in academic and professional settings.
7. Engage in lifelong learning and professional development.

Course Outcomes Bloom's

Course outcomes are statements that describe the knowledge, skills, and abilities that students should have acquired by the end of a course. These outcomes are often aligned with Bloom's Taxonomy, a framework for categorizing educational goals and objectives into different levels of complexity and specificity.

Bloom's Taxonomy consists of six levels:

1. **Remembering:** The ability to recall or retrieve previously learned information.
2. **Understanding:** The ability to comprehend the meaning of material.
3. **Applying:** The ability to use learned material in new and concrete situations.
4. **Analyzing:** The ability to break down material into its component parts so that its organizational structure may be understood.
5. **Evaluating:** The ability to make judgments about the value of ideas or materials.
6. **Creating:** The ability to put parts together to form a new whole.

By aligning course outcomes with Bloom's Taxonomy, instructors can ensure that their course is designed to help students achieve higher levels of thinking and learning. This can also help instructors to assess student learning and to design appropriate assessments and activities to support student achievement of the course outcomes.

Level

- A level is a tool used to determine if a surface is horizontal (level) or vertical (plumb).
- Levels can be used in construction, carpentry, surveying, and many other applications.
- There are several types of levels, including spirit levels, laser levels, and water levels.
- Spirit levels use a liquid-filled vial with an air bubble to indicate levelness.
- Laser levels project a laser beam to create a straight line on a surface.
- Water levels use the principle that water will always find its own level to determine levelness.
- Levels can vary in size and accuracy, with larger and more precise levels being used for more demanding applications.

- It is important to use a level when installing or building anything that needs to be level or plumb, such as shelves, cabinets, or walls.
- Using a level can help ensure that a project is completed accurately and to a high standard.

CO-1 Get an understanding of the use of different analytical instruments. K3

Analytical instruments are devices used to measure, analyze, and quantify the physical and chemical properties of a sample. These instruments are commonly used in scientific research, quality control, and medical diagnosis. Some of the most commonly used analytical instruments include:

1. **Spectrophotometers:** These instruments measure the absorption or transmission of light by a sample. They are commonly used to determine the concentration of a substance in a solution.
2. **Chromatographs:** These instruments separate and analyze the components of a mixture. They are commonly used to identify and quantify the components of a sample.
3. **Mass Spectrometers:** These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify the chemical composition of a sample.
4. **Electrochemical Instruments:** These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.
5. **Microscopes:** These instruments magnify small objects to make them visible to the naked eye. They are commonly used to study the structure of cells and tissues.

Understanding the use of these instruments is important for scientists and researchers to accurately analyze and interpret their data. It is also important for quality control and medical diagnosis to ensure accurate and reliable results.

CO-2 Measure the molecular / system properties such as surface tension

Surface tension is a property of liquids that arises due to the cohesive forces between the molecules at the surface of the liquid. These forces cause the liquid to behave as if it is covered by an elastic film, which resists deformation and tends to minimize the surface area of the liquid.

Some key points to remember when measuring surface tension are:

1. Surface tension is typically measured in units of force per unit length, such as millinewtons per meter (mN/m).

2. There are several methods for measuring surface tension, including the drop weight method, the maximum bubble pressure method, and the Wilhelmy plate method.
3. The choice of method will depend on the specific properties of the liquid being measured, as well as the desired accuracy and precision of the measurement.
4. It is important to ensure that the liquid is clean and free of contaminants, as these can affect the surface tension measurement.
5. Temperature can also affect surface tension, so it is important to control and record the temperature during the measurement.

Viscosity, Conductance of Solution, Chloride and Iron Content in the Water

- **Viscosity** is a measure of a fluid's resistance to flow. It describes the internal friction of a moving fluid. A fluid with high viscosity resists motion because its molecular makeup gives it a lot of internal friction. A fluid with low viscosity flows easily because its molecular makeup results in very little friction when it is in motion.
- **Conductance** is the ability of a solution to conduct an electric current. The more ions present in a solution, the higher its conductivity. High quality deionized water has a conductivity of about 0.05 S/cm at 25 °C, typical drinking water is in the range of 200–800 S/cm, while sea water is about 50 mS/cm (or 0.05 S/cm).
- **Chloride ions** are approximately 1.9 percent of the mass of seawater. The more chloride ions present in the water, the higher the conductivity. In general, the conductivity of waterways in the United States varies from 50 to 1500 $\mu\text{mhos/cm}$, and inland freshwater lake studies reveal a conductivity of 150 to 500 $\mu\text{mhos/cm}$.
- **Iron content** in water can affect its taste, color, and odor. It can also stain laundry and fixtures and promote the growth of certain bacteria. Iron is an essential element for human health, but high levels of iron in drinking water can have negative effects.

K3

- K3 is a term that can refer to several different things depending on the context.
- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a visa category for spouses of U.S. citizens.
- In mathematics, K3 could refer to a K3 surface, which is a type of algebraic surface.
- In music, K3 is a Belgian-Dutch girl group.

- Without more context, it is difficult to determine which specific meaning of K3 is being referred to.

CO-3 Measure the hardness and alkalinity of the water. K3

Hardness and alkalinity are two important parameters that are used to measure the quality of water. Hardness refers to the concentration of calcium and magnesium ions in water, while alkalinity refers to the ability of water to neutralize acids.

Here are the steps to measure the hardness and alkalinity of water:

1. **Collect a water sample:** Collect a water sample from the source you want to test. Make sure to use a clean container to avoid contamination.
2. **Measure the hardness:** There are several methods to measure the hardness of water, including titration, test strips, and electronic meters. Choose the method that is most suitable for your needs and follow the instructions provided by the manufacturer.
3. **Measure the alkalinity:** Alkalinity can be measured using a titration method. This involves adding a known amount of acid to the water sample and measuring the pH until it reaches a certain value. The amount of acid required to reach this value is used to calculate the alkalinity of the water.
4. **Record and interpret the results:** Record the results of the hardness and alkalinity tests and compare them to the recommended values for your specific application. High levels of hardness and alkalinity can affect the taste and appearance of water, as well as its ability to form scale and corrode pipes.

It is important to regularly measure the hardness and alkalinity of water to ensure its quality and suitability for its intended use.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an organic compound that is also known as carbolic acid. It is a white crystalline solid that is volatile and has a sweet, tarry odor. Phenol is an important industrial chemical that is used in the production of many products, including plastics, resins, and synthetic fibers.

There are several methods for the preparation of phenol, including:

1. **Cumene process:** This is the most widely used method for the production of phenol. In this process, cumene is oxidized to cumene hydroperoxide, which is then cleaved to produce phenol and acetone.
2. **Dow process:** In this process, chlorobenzene is reacted with dilute sodium hydroxide at high temperatures to produce phenol and sodium chloride.

3. **Raschig-Hooker process:** This process involves the reaction of benzene with sodium hydroxide and chlorine to produce phenol and sodium chloride.
4. **From coal tar:** Phenol can also be obtained from coal tar, which is a byproduct of the coking of coal. Coal tar contains a mixture of many organic compounds, including phenol, which can be separated and purified.

These are some of the fundamental concepts of the preparation of phenol. It is important to understand these concepts in order to have a better understanding of the industrial production of phenol and its uses.

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

- **Formaldehyde** is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.
- **Urea-formaldehyde (UF)**, also known as urea-methanal, is a nontransparent thermosetting resin or polymer. It is produced from urea and formaldehyde.
- UF is synthesized by the “alkali-acid-alkali” method using formaldehyde and urea as raw materials .
- Unlike most thermosetting resins, UF often shows the appearance of crystal domains .
- UF resins are used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- Melamine, polyvinyl alcohol, and adipic acid dihydrazide can be added to UF resin as modifiers for reducing the free formaldehyde.

K3

- K3 is a term that can refer to several different things, depending on the context in which it is used.
- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a visa category for spouses of U.S. citizens.
- In mathematics, K3 could refer to a complete graph with three vertices.
- In music, K3 is a Belgian girl band.
- In the context of technology, K3 could refer to a model of a device or product.

CO-5 Estimate the rate constant of reaction. K3

The rate constant of a reaction, denoted by the symbol k , is a measure of the speed of a chemical reaction. It is defined as the proportionality constant between the rate of the reaction and the concentrations of the reactants raised to their respective powers, as given by the rate law of the reaction.

To estimate the rate constant of a reaction, the following steps can be followed:

1. Determine the rate law of the reaction by conducting experiments to measure the initial rate of the reaction at different initial concentrations of the reactants.
2. Use the data obtained from the experiments to plot a graph of the initial rate of the reaction versus the initial concentration of the reactants.
3. From the graph, determine the order of the reaction with respect to each reactant.
4. Substitute the values of the order of the reaction and the initial concentrations of the reactants into the rate law to calculate the value of the rate constant, k .

It is important to note that the value of the rate constant, k , is dependent on the temperature at which the reaction is carried out. As the temperature increases, the rate constant also increases, leading to an increase in the rate of the reaction. The relationship between the rate constant and the temperature is given by the Arrhenius equation.

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ENGINEERING MATHEMATICS LAB KCS

Engineering Mathematics Lab is a course that is designed to provide students with hands-on experience in applying mathematical concepts and techniques to solve engineering problems. The course typically covers topics such as:

1. Numerical methods: This includes techniques for solving equations, interpolation, differentiation, and integration.
2. Linear algebra: This includes topics such as matrix operations, eigenvalues, and eigenvectors.
3. Probability and statistics: This includes topics such as probability distributions, hypothesis testing, and regression analysis.
4. Differential equations: This includes techniques for solving ordinary and partial differential equations.

The course is typically structured around a series of laboratory exercises that allow students to apply the mathematical concepts and techniques they have learned to solve practical engineering problems. The exercises may involve the use of mathematical software packages such as MATLAB or Mathematica.

The specific content and structure of the Engineering Mathematics Lab course

may vary depending on the institution and the specific engineering program. It is recommended that students consult with their academic advisor or the course instructor for more information about the specific requirements and expectations for the course.

Course Objectives:

- To provide students with a comprehensive understanding of the subject matter.
- To develop critical thinking and analytical skills.
- To enhance problem-solving abilities.
- To encourage independent learning and research.
- To prepare students for further studies or careers in the field.
- To promote effective communication and collaboration.
- To foster a sense of social responsibility and ethical behavior.

Fundamental Concepts of Analytical Instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are essential tools in various fields, including research, quality control, and environmental monitoring. Here are some fundamental concepts of analytical instruments:

1. **Accuracy and Precision:** Accuracy refers to how close a measurement is to the true value, while precision refers to how close repeated measurements are to each other. Analytical instruments must be both accurate and precise to provide reliable results.
2. **Sensitivity:** Sensitivity refers to the ability of an instrument to detect small changes in the property being measured. A highly sensitive instrument can detect even minute changes, making it useful for detecting trace amounts of substances.
3. **Selectivity:** Selectivity refers to the ability of an instrument to distinguish between different substances. A highly selective instrument can accurately identify and measure specific substances, even in the presence of other substances.
4. **Calibration:** Calibration is the process of adjusting an instrument to ensure that it provides accurate and precise measurements. Regular calibration is essential to maintain the reliability of analytical instruments.
5. **Signal-to-Noise Ratio:** The signal-to-noise ratio (SNR) is a measure of the quality of a measurement. A high SNR indicates that the signal (the measurement) is much stronger than the noise (background interference), resulting in a more reliable measurement.

These are some of the fundamental concepts of analytical instruments that students should understand. Understanding these concepts can help students use analytical instruments effectively and interpret their results accurately.

Analysis of Chloride Content, Hardness, and Alkalinity

1. Chloride content analysis is the process of measuring the amount of chloride ions present in a water sample. Chloride ions are negatively charged particles that can be found in water sources as a result of natural processes or human activities.
2. Hardness analysis measures the concentration of calcium and magnesium ions in water. These ions can cause water to be “hard,” which can lead to problems such as scaling in pipes and reduced effectiveness of soaps and detergents.
3. Alkalinity analysis measures the ability of water to neutralize acids. This is important because it helps to maintain a stable pH in the water, which is necessary for the health of aquatic life and for certain industrial processes.
4. These analyses are important for understanding the quality of water and for making informed decisions about its use and treatment. They can be performed using a variety of methods, including titration, colorimetry, and ion chromatography.
5. Understanding the results of these analyses can help to identify potential sources of pollution and to develop strategies for managing water quality. It can also help to ensure that water is safe for drinking, irrigation, and other uses.
6. In summary, the analysis of chloride content, hardness, and alkalinity is an important part of understanding and managing water quality. These analyses provide valuable information that can be used to make informed decisions about water use and treatment.

Water

Water is a transparent, tasteless, odorless, and nearly colorless chemical substance, which is the main constituent of Earth’s streams, lakes, and oceans, and the fluids of most living organisms. It is vital for all known forms of life, even though it provides no calories or organic nutrients.

- Water is a liquid at standard temperature and pressure.
- Water has a high specific heat capacity, meaning it can absorb a lot of heat before its temperature rises.

- Water is a good solvent, dissolving many substances to form aqueous solutions.
- Water is essential for life, with all known forms of life requiring it to survive.
- Water is involved in many chemical reactions, including hydrolysis and hydration reactions.
- Water exists in three states: solid (ice), liquid, and gas (water vapor).
- Water has a high surface tension, allowing it to form droplets and to support small objects on its surface.
- Water is a polar molecule, with a partial negative charge on the oxygen atom and partial positive charges on the hydrogen atoms.
- Water can form hydrogen bonds with other water molecules, leading to its high boiling point and other unique properties.
- Water is a renewable resource, with the water cycle constantly replenishing the Earth's supply of fresh water.

Measure of pH, Surface Tension and Viscosity

pH

- pH is a measure of the acidity or basicity of a solution.
- It is defined as the negative logarithm of the hydrogen ion concentration in a solution.
- The pH scale ranges from 0 to 14, with 7 being neutral.
- A pH value less than 7 indicates an acidic solution, while a pH value greater than 7 indicates a basic solution.

Surface Tension

- Surface tension is the property of a liquid that allows it to resist an external force.
- It is caused by the cohesive forces between the molecules of the liquid.
- Surface tension is measured in units of force per unit length, typically in millinewtons per meter (mN/m).
- The surface tension of a liquid can be measured using various methods, such as the drop weight method or the maximum bubble pressure method.

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.
- It is defined as the ratio of the shear stress to the shear rate in a fluid.
- Viscosity is typically measured in units of pascal-seconds ($\text{Pa} \cdot \text{s}$).
- The viscosity of a fluid can be measured using various methods, such as the capillary tube method or the rotating cylinder method.

Liquid

- A liquid is one of the four fundamental states of matter, along with solid, gas, and plasma.
- It is characterized by its ability to conform to the shape of its container while retaining a constant volume.
- Liquids have a definite volume but no fixed shape.
- The molecules in a liquid are in constant motion and are loosely packed, allowing them to slide past one another.
- The attractive forces between the molecules in a liquid are strong enough to keep the molecules close together but not strong enough to keep them in a fixed position.
- The temperature at which a substance changes from a solid to a liquid is called its melting point, and the temperature at which it changes from a liquid to a gas is called its boiling point.
- Liquids can be classified as either Newtonian or non-Newtonian, depending on whether their viscosity changes with the applied shear stress.
- Some common examples of liquids include water, milk, and oil.

Preparation of Different Resins

Resins can be derived from various sources such as polymers, monomers, oils, fatty acids, and natural sources . They have a wide range of applications including paints and coatings, personal care and home care, pharmaceuticals, rubber, plastics and their coatings, electrical and jewelry articles, decorative items, and wood coatings .

There are two categories of resinous products: natural resins and prepared resins . So far, no general method has been suggested or proposed for the preparation of resins .

Different types of resins include phenolic resins, alkyd resins, epoxy resins, UV resins, polyester resins, and polyurethane resins . The majority of resins are made up of two elements: the base resin and a catalyst (hardener). Combining the two elements results in a chemical reaction that allows the resin to set .

For example, by manipulating the structure of the linker, resins ranging from extremely acid labile to base labile can be prepared. Using linkers additionally allows the preparation of resins with special applications such as DHP resin utilized as a solid-phase support for alcohols or Weinreb resin utilized for preparing aldehydes and ketones .

Synthesis of Adipic Acid

Adipic acid is an important organic compound that is used in the production of nylon, polyurethane, and other polymers. Here are some key points to under-

stand about the synthesis of adipic acid:

1. Adipic acid is typically synthesized from cyclohexane or cyclohexanone through a process known as oxidation.
2. The oxidation of cyclohexane involves the use of a catalyst, typically cobalt or manganese, and an oxidizing agent, such as nitric acid or air.
3. The oxidation of cyclohexanone, on the other hand, can be achieved through the use of nitric acid or hydrogen peroxide.
4. Another method for the synthesis of adipic acid is through the carbonylation of butadiene, which involves the reaction of butadiene with carbon monoxide and water in the presence of a catalyst.
5. The choice of synthesis method depends on various factors, including the availability of starting materials, the desired yield, and the environmental impact of the process.

These are some of the key points to understand about the synthesis of adipic acid. It is an important process in the production of various polymers and has significant industrial applications.

Acid and Paracetamol by Conventional and Green Route

- Paracetamol, also known as acetaminophen, is a commonly used medication for pain relief and fever reduction.
- The conventional synthesis of paracetamol involves the nitration of phenol to produce p-nitrophenol, which is then reduced to p-aminophenol and acetylated to produce paracetamol.
- An alternative route for the synthesis of paracetamol using principles of green chemistry has been proposed. This involves solvent-free synthesis combined with microwave-energy irradiation .
- In 2014, Joncour et al., established this green approach involving direct amidation of hydroquinone to obtain paracetamol with maximum conversion which eliminates the problems associated with the earlier techniques such as, the extra step involved, unwanted ortho- isomerization, generation of the sulfate salt .
- Green chemistry aims to reduce the environmental impact of chemical processes by minimizing waste, reducing the use of hazardous substances, and increasing energy efficiency.
- The green synthesis of paracetamol is an example of how green chemistry principles can be applied to the production of commonly used medications.

List of Experiments

There are many experiments that have been conducted in various fields of science. Here are some examples:

1. **List Experiments:** A list experiment is a questionnaire design technique used to mitigate respondent social desirability bias when eliciting information about sensitive topics. With a large enough sample size, list experiments can be used to estimate the proportion of people for whom a sensitive statement is true.
2. **Lilo and Stitch Experiment Databank:** The official series of experiments, as stated by Jess Winfield, one of the executive producers, are as follows: 0-Series: Jumba's test batch, including many household helpers. 1-Series: Civic disturbances. 2-Series: Technological and scientific. 3-Series: Psychological. 4-Series: Mysterious series of mostly failed experiments.
3. **Harry Harlow's experiments:** Harry Harlow's experiments with baby monkeys and wire and cloth surrogate mothers (1957–1974).
4. **Stanley Milgram's experiments:** Stanley Milgram's experiments on human obedience (1963).
5. **Walter Mischel's marshmallow experiment:** Walter Mischel's marshmallow experiment showing the importance to life outcomes of the ability to delay gratification (beginning late 1960s).
6. **Science Experiments for Kids:** Science experiments you can do at home! Explore an ever growing list of hundreds of fun and easy science experiments. Have fun trying these experiments at home or use them for science fair project ideas.
7. **Science Buddies Science Experiments:** Fun science experiments to explore everything from kitchen chemistry to DIY mini drones. Easy to set up and perfect for home or school.

Calibration of Analytical Equipment and Apparatus

Calibration is the process of verifying the accuracy of a measuring instrument by comparing its readings with those of a known standard. This is an essential step in ensuring the accuracy and reliability of analytical results.

1. Calibration is necessary for all types of analytical equipment and apparatus, including balances, pipettes, thermometers, and spectrophotometers.
2. Calibration should be performed regularly, as specified by the manufacturer or according to a schedule established by the laboratory.
3. Calibration can be performed using certified reference materials, which are materials with known properties that have been certified by a recognized organization.
4. Calibration can also be performed using internal standards, which are materials with known properties that are prepared and used within the

laboratory.

5. Calibration records should be maintained, including the date of calibration, the results, and any corrective actions taken.
6. Calibration should be performed by trained personnel, following established procedures.
7. Calibration is an essential part of good laboratory practice and is necessary for ensuring the accuracy and reliability of analytical results.

Determination of Hardness of water sample by EDTA method

The hardness of water is due to the presence of calcium and magnesium salts in water. The EDTA method is one of the three main methods for determining the hardness of water. EDTA stands for Ethylenediaminetetraacetic acid. This reagent can form EDTA-metal complexes by reacting with metal ions, except for alkali metal ions.

In the EDTA method, water is titrated with ethylene diamine tetraacetic acid using Erichrome Black T (EBT) as an indicator. The procedure for calculating the hardness of water by EDTA titration is as follows:

1. Take a sample volume of 20ml (V ml).
2. Dilute 20ml of the sample in an Erlenmeyer flask to 40ml by adding 20ml of distilled water.
3. Add 1 mL of ammonia buffer to bring the pH to 10 ± 0.1 .
4. Add 1 or 2 drops of the indicator solution.
5. Add EDTA titrant to the sample with vigorous shaking until the wine red color just turns blue.

The hardness can then be calculated as mg/L using the formula: Hardness (EDTA), as mg/L = $\frac{A}{S} \times T \times 1,000$ where A = mL of EDTA titrant used, T = Titer of EDTA titrant, mg CaCO₃ per mL of EDTA titrant, and S = mL of sample volume. The result can be reported as "Total Hardness (EDTA) = _____ mg/L as CaCO₃" or as "Calcium Hardness (EDTA) = _____ mg/L as CaCO₃".

3. Determination of Alkalinity of Water Sample

Alkalinity is a measure of the ability of water to neutralize acids. It is an important parameter in determining the quality of water for various uses. Here are the steps to determine the alkalinity of a water sample:

1. Collect a water sample in a clean container.
2. Measure the pH of the water sample using a pH meter or pH test strips.
3. If the pH is above 7, the water is alkaline.
4. To determine the exact alkalinity, titrate the water sample with a standard acid solution, such as sulfuric acid or hydrochloric acid.

5. The volume of acid required to bring the pH of the water sample to a specific value, usually 4.5, is used to calculate the alkalinity of the water sample.
6. The alkalinity is usually expressed in milligrams per liter (mg/L) of calcium carbonate (CaCO_3).

These are the basic steps to determine the alkalinity of a water sample. It is important to follow proper procedures and use accurate equipment to obtain reliable results.

Determination of pH by Titrimetric Method

Titration is a laboratory technique used to determine the concentration of an unknown solution by reacting it with a solution of known concentration. The titrimetric method can be used to determine the pH of a solution.

1. **Principle:** The principle behind the titrimetric method for determining pH is the neutralization reaction between an acid and a base. A solution of known concentration, either an acid or a base, is added to the solution being tested until the reaction is complete. The volume of the known solution required to reach the endpoint of the reaction is used to calculate the concentration of the unknown solution, and thus its pH.
2. **Procedure:** The procedure for determining pH by titration involves the following steps:
 - A solution of known concentration, either an acid or a base, is prepared.
 - A sample of the solution being tested is placed in a flask or beaker.
 - An indicator is added to the solution being tested. The indicator changes color at the endpoint of the reaction.
 - The solution of known concentration is added to the solution being tested, drop by drop, while stirring.
 - The addition of the known solution is continued until the indicator changes color, indicating that the endpoint of the reaction has been reached.
 - The volume of the known solution required to reach the endpoint is recorded.
 - The concentration of the unknown solution is calculated using the volume of the known solution and the balanced chemical equation for the reaction.
3. **Calculations:** The concentration of the unknown solution can be calculated using the formula: $C_1V_1 = C_2V_2$, where C_1 is the concentration of the known solution, V_1 is the volume of the known solution, C_2 is the concentration of the unknown solution, and V_2 is the volume of the unknown solution. Once the concentration of the unknown solution is known, its

pH can be calculated using the formula: $\text{pH} = -\log[\text{H}^+]$, where $[\text{H}^+]$ is the concentration of hydrogen ions in the solution.

4. **Advantages:** The titrimetric method for determining pH is simple, accurate, and precise. It can be used to determine the pH of a wide range of solutions, including those that are colored or turbid.
5. **Limitations:** The titrimetric method for determining pH requires the use of an indicator, which may not be suitable for all solutions. The method also requires the preparation of a solution of known concentration, which may not always be available or practical.

Determination of Surface Tension of Given Liquid

Surface tension is the property of a liquid that allows it to resist an external force. It is the result of cohesive forces between the molecules of the liquid. There are several methods to determine the surface tension of a given liquid, including the following:

1. **Maximum Bubble Pressure Method:** This method involves measuring the maximum pressure required to blow a bubble from a submerged tube. The surface tension is calculated from the pressure difference between the inside and outside of the bubble.
2. **Drop Weight Method:** This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube. The surface tension is calculated from the weight of the droplet and the radius of the tube.
3. **Du Nouy Ring Method:** This method involves measuring the force required to detach a ring from the surface of a liquid. The surface tension is calculated from the force and the circumference of the ring.
4. **Wilhelmy Plate Method:** This method involves measuring the force required to detach a flat plate from the surface of a liquid. The surface tension is calculated from the force and the perimeter of the plate.
5. **Capillary Rise Method:** This method involves measuring the height to which a liquid rises in a thin tube due to capillary action. The surface tension is calculated from the height of the liquid, the radius of the tube, and the density of the liquid.

Each of these methods has its own advantages and limitations, and the choice of method depends on the specific requirements of the experiment. It is important to carefully calibrate the instruments and follow the appropriate procedures to obtain accurate results.

Determination of Viscosity of a given liquid by viscometer

Viscosity is a measure of a fluid's resistance to flow. It is commonly measured using a device called a viscometer. Here are the steps to determine the viscosity of a given liquid using a viscometer:

1. **Select the appropriate viscometer:** Different types of viscometers are available for different types of liquids and measurement ranges. Choose the one that is suitable for the liquid you want to measure.
2. **Calibrate the viscometer:** Before using the viscometer, it must be calibrated to ensure accurate measurements. Follow the manufacturer's instructions to calibrate the device.
3. **Prepare the sample:** The liquid to be measured should be free of air bubbles and at a consistent temperature. Pour the liquid into the viscometer, taking care not to introduce air bubbles.
4. **Measure the flow time:** Start the timer when the liquid begins to flow through the viscometer and stop it when the liquid reaches the marked line. Record the time it takes for the liquid to flow through the viscometer.
5. **Calculate the viscosity:** Use the flow time and the calibration constant of the viscometer to calculate the viscosity of the liquid. The calculation method may vary depending on the type of viscometer used.
6. **Repeat the measurement:** To ensure accuracy, repeat the measurement several times and take the average of the results.

These are the basic steps to determine the viscosity of a given liquid using a viscometer. It is important to follow the manufacturer's instructions and standard procedures to ensure accurate and reliable results.

Determination of the strength of Ferrous ammonium sulfate using external indicator

1. Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate.
2. Its chemical formula is $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$.
3. It is commonly used as a standard solution in redox titrations due to its stability and the ease of preparation.
4. The strength of a ferrous ammonium sulfate solution can be determined by titrating it against a standard solution of potassium permanganate (KMnO_4) using an external indicator.
5. The external indicator used in this titration is potassium ferricyanide ($\text{K}_3\text{Fe}(\text{CN})_6$).

6. The reaction between ferrous ammonium sulfate and potassium permanganate is a redox reaction, where the iron in ferrous ammonium sulfate is oxidized from Fe^{2+} to Fe^{3+} and the manganese in potassium permanganate is reduced from Mn^{7+} to Mn^{2+} .
7. The end point of the titration is indicated by the disappearance of the blue color of the potassium ferricyanide solution, which indicates that all the Fe^{2+} ions have been oxidized to Fe^{3+} ions.
8. The strength of the ferrous ammonium sulfate solution can be calculated using the volume of potassium permanganate solution used and the molarity of the potassium permanganate solution.

Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is a primary standard substance that can be used to determine the strength of a solution.
2. An internal indicator is used to determine the end point of the titration.
3. The internal indicator used in this experiment is potassium iodide.
4. The reaction between potassium dichromate and potassium iodide produces iodine, which can be detected by its characteristic color.
5. The end point of the titration is reached when all the iodine has been consumed and the solution turns from a yellow color to a colorless solution.
6. The strength of the potassium dichromate solution can be calculated using the volume of the solution used and the concentration of the iodine solution.
7. This method is accurate and reliable for determining the strength of potassium dichromate solutions.

9. Determination of available chlorine in bleaching powder

The available chlorine present in a bleaching powder sample can be determined iodometrically by treating its solution with an excess of potassium iodide solution in an acidic medium. The liberated iodine (I_2) is then treated with sodium thiosulphate ($\text{Na}_2\text{S}_2\text{O}_3$) solution using freshly prepared starch solution as an indicator to be added near the endpoint.

This process is known as iodometry, and the calculation of available chlorine is called iodometric calculation. A sample of bleaching powder is taken, and when treated with dilute acid or water, it liberates chlorine gas.

The available chlorine in the given sample of bleaching powder can be determined by the iodometric method. Bleaching powder is commonly used as a disinfectant, and the chlorine present in the bleaching powder gets reduced

with time.

Determination of Chloride Content in Water Sample

1. Chloride ions are one of the major inorganic anions in water and wastewater.
2. The chloride test measures the level of chloride ions in water.
3. The presence of chloride ions in water can be an indicator of pollution from human or animal waste, industrial processes, or saltwater intrusion.
4. The most common method for determining the chloride content in water is the argentometric method, also known as the Mohr method.
5. In this method, a measured volume of water sample is titrated with a standard solution of silver nitrate using potassium chromate as an indicator.
6. The end point of the titration is indicated by the formation of a red-brown precipitate of silver chromate.
7. The amount of silver nitrate used in the titration is proportional to the chloride content in the water sample.
8. The chloride content can be calculated using the formula: $\text{Cl}^- (\text{mg/L}) = (\text{AgNO}_3 \text{ used (mL)} \times \text{AgNO}_3 \text{ normality} \times 35.45) / \text{sample volume (mL)}$.
9. Other methods for determining the chloride content in water include ion chromatography, ion-selective electrode, and spectrophotometry.
10. It is important to regularly monitor the chloride content in water to ensure that it is within safe levels for human consumption and for the protection of aquatic life.

11. Preparation of Phenol formaldehyde (PF) resin

Phenol formaldehyde (PF) resin, also known as phenolic resin, is a synthetic polymer obtained by the reaction of phenol with formaldehyde. Here are the steps involved in the preparation of PF resin:

1. **Phenol and formaldehyde** are mixed in a reactor vessel in the presence of an acidic or basic catalyst.
2. The mixture is heated and stirred to initiate the **polymerization reaction**.
3. The reaction is allowed to proceed until the desired degree of polymerization is achieved.
4. The resulting resin is then cooled and solidified.
5. The solid resin is then ground into a fine powder or processed into the desired shape.

The properties of the PF resin can be controlled by varying the ratio of phenol

to formaldehyde, the type of catalyst used, and the reaction conditions. PF resins are widely used in the production of laminates, adhesives, and molded products due to their high strength, heat resistance, and chemical resistance.

12. Preparation of Urea Formaldehyde (UF) Resin

Urea Formaldehyde (UF) resin is a type of thermosetting polymer that is commonly used in the production of particleboard, medium-density fiberboard (MDF), and other wood products. It is made by reacting urea and formaldehyde in the presence of a catalyst.

Here are the steps involved in the preparation of UF resin:

1. **Mixing of reactants:** Urea and formaldehyde are mixed in a reactor vessel in a molar ratio of 1:1.5 to 2.5. The pH of the mixture is adjusted to 7.5-8.5 using an alkaline catalyst such as sodium hydroxide or ammonia.
2. **Condensation reaction:** The mixture is heated to 70-90°C and stirred for 1-2 hours to initiate the condensation reaction. During this stage, methylene and dimethylene ether linkages are formed between the urea and formaldehyde molecules.
3. **Addition of modifiers:** Modifiers such as urea, melamine, or ammonium chloride may be added to the reaction mixture to control the properties of the final resin. These modifiers can affect the curing time, water resistance, and other properties of the UF resin.
4. **Cooling and storage:** The reaction mixture is cooled to room temperature and stored in a container until it is ready to be used. The resin can be stored for several weeks before it begins to lose its reactivity.

These are the basic steps involved in the preparation of UF resin. The exact process may vary depending on the specific application and desired properties of the final product.

Preparation of Adipic acid

Adipic acid is an organic compound with the formula $(\text{CH}_2)_4(\text{COOH})_2$. It is the most important dicarboxylic acid from an industrial perspective, with about 2.5 billion kilograms of this white crystalline powder being produced annually, mainly as a precursor for the production of nylon .

Adipic acid is produced from a mixture of cyclohexanone and cyclohexanol called KA oil, the abbreviation of ketone-alcohol oil. The KA oil is oxidized with nitric acid to give adipic acid, via a multistep pathway .

Preparation of Paracetamol

Paracetamol is prepared from p-aminophenol by acetylating it with acetic anhydride in the presence of 3-4 drops of concentrated sulphuric acid as a catalyst .

Paracetamol is available as 325-mg or 500-mg tablets, which may include calcium stearate or magnesium stearate, cellulose, docusate sodium and sodium benzoate or sodium lauryl sulfate, starch, hydroxypropyl methylcellulose, propylene glycol, sodium starch glycolate, polyethylene glycol and Red #40 .

Determination of Cell Conductance of a Solution

1. Conductance is the measure of the ability of a solution to conduct an electric current.
2. The conductance of a solution is determined by the concentration and mobility of the ions present in the solution.
3. The cell conductance of a solution is the conductance of the solution measured between two electrodes of a conductivity cell.
4. The cell constant is the ratio of the distance between the electrodes to the area of the electrodes.
5. The cell conductance is calculated by multiplying the measured conductance by the cell constant.
6. The cell conductance can be determined using a conductivity meter, which measures the conductance of the solution and automatically applies the cell constant to calculate the cell conductance.
7. The cell conductance can also be determined using a Wheatstone bridge circuit, where the unknown conductance of the solution is compared to a known conductance to determine the cell conductance.
8. The cell conductance is affected by temperature, and temperature compensation is necessary to obtain accurate measurements.
9. The cell conductance can be used to determine the concentration of ions in a solution, by comparing the measured cell conductance to a calibration curve obtained using solutions of known concentrations.

Determination of Rate constant of hydrolysis of esters

1. The rate constant of hydrolysis of esters can be determined by measuring the rate of reaction at different concentrations of the reactants.
2. The reaction between an ester and water to form a carboxylic acid and an alcohol is known as hydrolysis.
3. The rate of hydrolysis of esters is influenced by several factors, including the nature of the ester, the temperature, and the pH of the solution.

4. The rate law for the hydrolysis of esters can be expressed as: $\text{rate} = k[\text{ester}]$, where k is the rate constant, $[\text{ester}]$ is the concentration of the ester, and water is the concentration of water.
5. The rate constant, k , can be determined by measuring the initial rate of the reaction at different concentrations of the ester and water, and then using the rate law to calculate the value of k .
6. The initial rate of the reaction can be determined by measuring the concentration of one of the products, such as the carboxylic acid, at different times after the start of the reaction.
7. The concentration of the carboxylic acid can be measured using a suitable analytical technique, such as titration or spectrophotometry.
8. Once the initial rate of the reaction has been determined at different concentrations of the ester and water, the data can be plotted and the rate constant, k , can be calculated from the slope of the line.
9. The value of the rate constant, k , is specific to the particular ester being studied and the conditions under which the reaction is carried out.
10. The rate constant, k , can be used to predict the rate of the hydrolysis reaction under different conditions, such as at different temperatures or in the presence of a catalyst.
11. The determination of the rate constant of hydrolysis of esters is an important tool in the study of chemical kinetics and the mechanisms of chemical reactions.
12. The rate constant of hydrolysis of esters can also be used in the design of chemical processes and the development of new chemical products.
13. The hydrolysis of esters is an important reaction in many industrial processes, including the production of soaps, detergents, and biodiesel.
14. The determination of the rate constant of hydrolysis of esters is therefore of great practical importance in the chemical industry.
15. The study of the hydrolysis of esters also provides valuable insights into the fundamental principles of chemical reactivity and the nature of chemical bonds.

Element detection and identification of functional groups in organic compounds

1. **Element detection** refers to the process of identifying the presence of specific elements in a given compound.
2. **Functional groups** are specific groups of atoms within a molecule that are responsible for the characteristic chemical reactions of that molecule.
3. There are several methods for detecting elements and identifying functional groups in organic compounds, including:
 - **Combustion analysis:** This method involves burning a sample of the compound and analyzing the resulting products to determine the presence of elements such as carbon, hydrogen, and nitrogen.

- **Infrared spectroscopy:** This technique uses infrared radiation to identify the presence of specific functional groups based on their characteristic absorption patterns.
 - **Mass spectrometry:** This method involves ionizing a sample of the compound and analyzing the resulting fragments to determine the molecular weight and structure of the compound, which can provide information about the presence of specific elements and functional groups.
 - **Nuclear magnetic resonance spectroscopy:** This technique uses the interaction of nuclear spins with a magnetic field to provide information about the chemical environment of specific atoms within a molecule, which can be used to identify the presence of specific functional groups.
4. These methods can be used individually or in combination to provide a comprehensive analysis of the elements and functional groups present in a given organic compound.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose any 10 experiments from the list provided.
- The instructor also has the option to change any two of the experiments from the list.
- This allows the instructor to tailor the experiments to the needs and interests of the students.
- It is important for students to carefully read the list of experiments and be prepared for any changes made by the instructor.
- Students should communicate with the instructor to clarify any questions or concerns about the experiments chosen.

Course Outcomes

Course outcomes are statements that describe the knowledge, skills, and abilities that students should possess after completing a course. These outcomes serve as a guide for instructors to design course content, assessments, and learning activities. They also help students understand the purpose of the course and what they can expect to learn.

Some key points to consider when developing course outcomes include:

1. Outcomes should be specific, measurable, and achievable.
2. Outcomes should align with the overall goals of the program or curriculum.

3. Outcomes should be written in clear and concise language.
4. Outcomes should focus on what students will be able to do, rather than what they will know.
5. Outcomes should be reviewed and updated regularly to ensure they remain relevant and accurate.

By clearly defining course outcomes, instructors can create a more effective and engaging learning experience for students. Additionally, course outcomes can help students take ownership of their learning and better understand how the course fits into their overall educational goals.

Upon completion of the course, the student should be able to:

1. Demonstrate a thorough understanding of the course material and its key concepts.
2. Apply the knowledge and skills acquired during the course to real-world situations.
3. Analyze and evaluate information critically and effectively.
4. Communicate ideas and arguments clearly and effectively, both orally and in writing.
5. Work collaboratively with others to achieve common goals.
6. Demonstrate the ability to think creatively and independently.
7. Demonstrate ethical and responsible behavior in academic and professional settings.
8. Engage in lifelong learning and professional development.

Course Outcomes Bloom's

Course outcomes are statements that describe the knowledge, skills, and abilities that students should have acquired by the end of a course. These outcomes are typically aligned with the course objectives and are used to assess the effectiveness of the course in achieving its goals.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. It is often used to design course outcomes and assessments. The taxonomy consists of six levels, arranged in a hierarchy from lower-order thinking skills to higher-order thinking skills:

1. **Remembering:** The ability to recall or retrieve previously learned information.
2. **Understanding:** The ability to comprehend the meaning of material.
3. **Applying:** The ability to use learned material in new and concrete situations.
4. **Analyzing:** The ability to break down material into its component parts so that its organizational structure may be understood.
5. **Evaluating:** The ability to make judgments about the value of ideas or materials.

6. **Creating:** The ability to put parts together to form a new whole.

By using Bloom's Taxonomy to design course outcomes, instructors can ensure that their courses are challenging and engaging for students, and that assessments accurately measure student learning. Course outcomes written using Bloom's Taxonomy are specific, measurable, and aligned with the course objectives. They provide a clear roadmap for students to follow as they progress through the course, and help instructors to design effective assessments that accurately measure student learning.

Level

- A level is a tool used to determine if a surface is horizontal or vertical.
- It consists of a small glass tube filled with liquid, usually alcohol or a colored spirit, with an air bubble inside.
- When the level is placed on a surface, the bubble will move to the highest point of the tube, indicating if the surface is level or not.
- There are different types of levels, including spirit levels, laser levels, and digital levels.
- Levels are commonly used in construction, carpentry, and surveying, among other fields.
- It is important to use a level when installing or building structures to ensure they are straight and level, which can affect their stability and appearance.

CO-1: Understanding the use of different analytical instruments (K3)

Analytical instruments are tools used to measure and analyze physical, chemical, and biological properties of substances. These instruments are used in a variety of fields, including research, quality control, and environmental monitoring.

Some common analytical instruments include:

1. **Spectrophotometers:** These instruments measure the absorption or transmission of light by a sample. They are commonly used to determine the concentration of a substance in a solution.
2. **Chromatographs:** These instruments separate mixtures into their individual components. They are commonly used to identify and quantify the components of a mixture.
3. **Mass spectrometers:** These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify unknown compounds and to determine the elemental composition of a sample.

4. **Electrochemical instruments:** These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.

Each of these instruments has its own unique capabilities and limitations. Understanding the use of these instruments is essential for selecting the appropriate tool for a given analytical task.

CO₂: Measuring Molecular/System Properties such as Surface Tension

Carbon dioxide (CO₂) is a colorless, odorless gas that is naturally present in the Earth's atmosphere. It is an important greenhouse gas that helps regulate the temperature of the planet. However, human activities such as burning fossil fuels and deforestation have significantly increased the concentration of CO₂ in the atmosphere, contributing to global warming and climate change.

Surface tension is a property of liquids that arises from the cohesive forces between molecules at the surface of the liquid. It is the force that allows insects to walk on water and causes water droplets to form beads on a surface.

To measure the surface tension of CO₂, several methods can be used, including:

1. **Maximum Bubble Pressure Method:** This method involves measuring the maximum pressure required to force a gas bubble through a liquid.
2. **Drop Weight Method:** This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube.
3. **Wilhelmy Plate Method:** This method involves measuring the force required to detach a thin plate from the surface of a liquid.

These methods can be used to measure the surface tension of CO₂ and other gases, providing valuable information about their molecular and system properties. It is important to note that the surface tension of CO₂ can vary depending on factors such as temperature and pressure, so these variables must be carefully controlled during measurement.

Viscosity, Conductance of Solution, Chloride and Iron Content in Water

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.
- The viscosity of formation water is a function of pressure, temperature, and dissolved solids.
- In general, brine viscosity increases with increasing pressure, increasing salinity, and decreasing temperature.

- Dissolved gas in the formation water at reservoir conditions generally results in a negligible effect on water viscosity.

Conductance of Solution

- Conductivity is a measure of a solution's ability to conduct an electric current.
- High-quality deionized water has a conductivity of about 0.05 S/cm at 25 °C, typical drinking water is in the range of 200–800 S/cm, while seawater is about 50 mS/cm (or 0.05 S/cm).
- The more chloride ions present in the water, the higher the conductivity.
- In general, the conductivity of waterways in the United States varies from 50 to 1500 μ mhos/cm, and inland freshwater lake studies reveal a conductivity of 150 to 500 μ mhos/cm.

Chloride Content in Water

- Chloride ions are approximately 1.9 percent of the mass of seawater.
- The more chloride ions present in the water, the higher the conductivity.

Iron Content in Water

- Iron content in water is not mentioned in the search results.

K3

- K3 is a term that can refer to several different things, depending on the context in which it is used.
- In mathematics, K3 refers to a K3 surface, which is a complex algebraic surface with trivial canonical bundle.
- In mountaineering, K3 is the name given to the mountain Skil Brum, which is located in the Karakoram mountain range.
- In music, K3 is a Belgian-Dutch girl group that was formed in 1998.
- In biology, K3 is a type of edentulous jaw relation recorder used in dentistry.
- In transportation, K3 may refer to a model of passenger car produced by the South Korean automaker Kia Motors.

CO-3 Measure the hardness and alkalinity of the water. K3

Hardness and alkalinity are two important parameters of water quality. They are used to determine the suitability of water for various uses, such as drinking, irrigation, and industrial processes.

1. **Hardness** refers to the concentration of calcium and magnesium ions in water. These ions can form insoluble compounds with soap, causing it to lose its cleaning power. Hard water can also cause scaling in pipes and appliances, reducing their efficiency and lifespan.
2. **Alkalinity** refers to the ability of water to neutralize acids. It is a measure of the buffering capacity of water, which is determined by the presence of bicarbonate, carbonate, and hydroxide ions. Alkalinity is important for maintaining a stable pH in aquatic ecosystems and for preventing corrosion in pipes and appliances.

There are several methods for measuring the hardness and alkalinity of water. These include:

- **Titration:** This involves adding a chemical reagent to a water sample until a color change occurs, indicating that all the ions of interest have reacted. The amount of reagent used is proportional to the concentration of ions in the water.
- **Colorimetry:** This method uses a colorimetric reagent that reacts with the ions of interest to produce a color change. The intensity of the color is proportional to the concentration of ions in the water.
- **Ion chromatography:** This is a more advanced method that separates and measures the individual ions in a water sample using a specialized instrument.

It is important to measure the hardness and alkalinity of water regularly to ensure that it is suitable for its intended use and to prevent problems such as scaling and corrosion. These measurements can also provide valuable information for managing water resources and protecting aquatic ecosystems.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an organic compound that can be prepared through several methods. Some of the fundamental concepts of the preparation of phenol are:

1. **Preparation of Phenols from Haloarenes:** Chlorobenzene is an example of a haloarene which is formed by the monosubstitution of the benzene ring. When chlorobenzene is fused with sodium hydroxide at 623K and 320 atm, sodium phenoxide is produced .
2. **Preparation of Phenols from Diazonium Salts:** Phenols can also be prepared from diazonium salts.
3. **Preparation of Phenols from Benzene Sulphonic Acid:** At 573 – 623K, sodium phenoxide is produced by fusing the sodium salt of benzene

sulphonic acid with sodium hydroxide. Sodium phenoxide is converted to phenol when it is treated with dilute acid or carbon dioxide.

4. **Industrial Preparation of Phenols:** Phenol can be prepared industrially through several methods such as extraction from coal tar, Raschig's Process, and the air oxidation of cumene.
5. **Synthesis of Phenols:** Phenols can be synthesized in large quantities by the pyrolysis of the sodium salt of benzene sulfonic acid, by the Dow process, and by the air oxidation of cumene. Small amounts of phenol can also be prepared by the peroxide oxidation of phenylboronic acid and the hydrolysis of diazonium salts.

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

Formaldehyde

- Formaldehyde is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.

Urea Formaldehyde Resin

- Urea-formaldehyde (UF) is a nontransparent thermosetting resin or polymer.
- It is produced from urea and formaldehyde.
- These resins are used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- A stable urea-formaldehyde resin (UF) is synthesized by the "alkali-acid-alkali" method.
- Unlike most thermosetting resins, UF often shows the appearance of crystal domains.

Adipic Acid

- Melamine, polyvinyl alcohol, and adipic acid dihydrazide as modifiers were added to urea-formaldehyde (UF) resin for reducing the free formaldehyde.

Paracetamol

- No information was found on Paracetamol in the search results.

K3

- K3 is a synthetic, artificially produced form of vitamin K that doesn't occur naturally.

- This is unlike the other two forms of vitamin K — vitamin K1, known as phylloquinone, and vitamin K2, called menaquinone.
- Vitamin K3 may be converted into K2 in your liver.
- Vitamin K3 is also known as menadione.
- Vitamin K is important for blood clotting and bone health.
- Although vitamin K3 has demonstrated anticancer and antibacterial properties in test-tube studies, it has potential side effects.
- K3 is also the name of a form used by the Internal Revenue Service (IRS) in the United States. It is called Schedule K-3 (Form 1065) and is used to report a partner's share of income, deductions, credits, etc. for international tax purposes.

CO-5 Estimate the rate constant of reaction. K3

- The rate constant of a reaction, denoted by k , is a proportionality constant that relates the rate of a chemical reaction to the concentrations of the reactants.
- The rate constant is specific to a particular reaction and is determined experimentally.
- The rate constant can be calculated by rearranging the rate law equation. For example, for a reaction with the rate law $\text{Rate} = k [A]^a [B]^b$, the rate constant can be calculated as $k = \text{Rate} / ([A]^a [B]^b)$.
- For a termolecular reaction, the reaction rate is described by $r = k_3 [A] [B] [C]$, where k_3 is a termolecular rate constant.
- The equilibrium constant for a reaction that is the sum of two or more reactions is equal to the product of the equilibrium constants for the individual reactions.

Page 32 of 40

- Page 32 is the thirty-second page of a forty-page document or book.
- It is located after page 31 and before page 33.
- The content of page 32 will vary depending on the specific document or book it is a part of.
- It is important to read and understand the content of page 32 in the context of the entire document or book.
- Page 32 may contain text, images, diagrams, or a combination of these elements.
- The information on page 32 may be important for understanding the overall message or argument of the document or book.
- It is recommended to take notes or highlight key points while reading page 32 to aid in comprehension and retention of the information presented.

ENGINEERING MATHEMATICS LAB

Engineering Mathematics Lab is a course designed to provide students with hands-on experience in applying mathematical concepts and techniques to solve engineering problems. The course typically covers topics such as:

1. **Differential Equations:** Students learn to solve first and second order differential equations, and apply these techniques to model and analyze engineering systems.
2. **Linear Algebra:** Students learn to perform matrix operations, find eigenvalues and eigenvectors, and use these concepts to solve systems of linear equations.
3. **Numerical Methods:** Students learn to use numerical methods such as Newton's method, the bisection method, and the secant method to find roots of equations, and to use numerical integration and differentiation techniques to approximate solutions to differential equations.
4. **Probability and Statistics:** Students learn to calculate probabilities, find expected values, and use statistical methods to analyze data and make predictions.
5. **Fourier Series and Transforms:** Students learn to represent periodic functions as sums of sines and cosines, and to use Fourier transforms to analyze signals and systems.

The course typically includes both lectures and laboratory sessions, where students work on problems and projects to apply the concepts they have learned. The course is designed to help students develop problem-solving skills and to prepare them for more advanced courses in engineering and applied mathematics.

Course Objectives:

1. To provide a comprehensive understanding of the subject matter.
2. To develop critical thinking and analytical skills.
3. To enhance the ability to apply theoretical concepts to real-world scenarios.
4. To encourage independent learning and research.
5. To foster effective communication and collaboration skills.
6. To prepare students for further academic or professional pursuits in the field.
7. To promote ethical and responsible behavior in the discipline.

1. To enable the students to understand about the fundamental concepts of analytical instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are commonly used in scientific research, medical diagnosis, and industrial processes.

Some fundamental concepts of analytical instruments include:

- **Accuracy:** The degree to which the measurement result agrees with the true value of the quantity being measured.
- **Precision:** The degree to which repeated measurements of the same quantity agree with each other.
- **Sensitivity:** The smallest change in the quantity being measured that can be detected by the instrument.
- **Resolution:** The smallest difference between two measurements that can be distinguished by the instrument.
- **Calibration:** The process of adjusting the instrument to ensure that its measurements are accurate.
- **Signal-to-noise ratio:** The ratio of the strength of the signal being measured to the background noise.

Understanding these concepts is essential for the proper use and interpretation of analytical instruments. It is important for students to learn about these concepts in order to effectively use analytical instruments in their future studies and careers.

2. Analysis of Chloride Content, Hardness, and Alkalinity of Water

- **Chloride Content:** Chloride is one of the major anions found in water and wastewater. The chloride content of water can be determined by titration with silver nitrate using potassium chromate as an indicator. This method is known as the Mohr method.
- **Hardness:** Hardness in water is caused by the presence of dissolved calcium and magnesium ions. It is usually expressed in terms of calcium carbonate. Hardness can be determined by titration with a standard solution of ethylenediaminetetraacetic acid (EDTA) using eriochrome black T as an indicator.
- **Alkalinity:** Alkalinity is a measure of the capacity of water to neutralize acids. It is mainly caused by the presence of bicarbonate, carbonate, and hydroxide ions. Alkalinity can be determined by titration with a standard solution of sulfuric acid using phenolphthalein and/or methyl orange as an indicator.

These are some of the basic concepts and methods used in the analysis of chloride content, hardness, and alkalinity of water. Understanding these concepts and methods is important for students studying water chemistry and treatment.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of a liquid.

- **pH:** pH is a measure of the acidity or basicity of a solution. It is defined as the negative logarithm of the hydrogen ion concentration. The pH scale ranges from 0 to 14, with 7 being neutral. A pH less than 7 indicates an acidic solution, while a pH greater than 7 indicates a basic solution. pH can be measured using a pH meter or pH paper.
- **Surface tension:** Surface tension is the property of a liquid that allows it to resist an external force. It is caused by the cohesive forces between the liquid molecules. Surface tension can be measured using a number of methods, including the drop weight method, the maximum bubble pressure method, and the Du Nouy ring method.
- **Viscosity:** Viscosity is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate. Viscosity can be measured using a number of methods, including the capillary tube method, the falling ball method, and the rotational viscometer method. Viscosity is affected by temperature, with most fluids becoming less viscous as the temperature increases.

4. To enable the students to understand about the preparation of different resins.

Resins are solid or highly viscous substances that are typically convertible into polymers. They are usually mixtures of organic compounds, mainly terpenes and their derivatives. Here are some points to consider when preparing different resins:

1. **Types of resins:** There are various types of resins, including natural resins, synthetic resins, and modified natural resins. Each type of resin has its own unique properties and preparation methods.
2. **Raw materials:** The raw materials used in the preparation of resins vary depending on the type of resin being prepared. For example, natural resins are obtained from plants, while synthetic resins are prepared from chemicals.
3. **Preparation methods:** The methods used to prepare resins vary depending on the type of resin being prepared. Some common methods include polymerization, condensation, and addition reactions.
4. **Safety precautions:** The preparation of resins involves the use of chemicals and heat, so it is important to follow safety precautions to prevent accidents. This includes wearing protective clothing and using proper ventilation.
5. **Quality control:** To ensure that the resins being prepared meet the desired specifications, it is important to implement quality control measures.

This includes testing the resins for properties such as viscosity, hardness, and color.

In summary, the preparation of different resins involves understanding the types of resins, the raw materials used, the preparation methods, safety precautions, and quality control measures. By following these guidelines, students can learn how to prepare different resins effectively.

5. To enable the students to understand about the synthesis of organic compounds such as adipic acid and paracetamol by conventional and green route.

Adipic acid and paracetamol are two important organic compounds that can be synthesized through both conventional and green routes. The conventional route involves the use of traditional chemical methods, while the green route involves the use of more environmentally friendly methods.

1. **Adipic Acid:** Adipic acid is an important dicarboxylic acid that is used in the production of nylon and other polymers. It can be synthesized through the conventional route by the oxidation of cyclohexanol or cyclohexanone with nitric acid. This process produces large amounts of nitrous oxide, a potent greenhouse gas. The green route for the synthesis of adipic acid involves the use of bio-based methods, such as the fermentation of sugars or the oxidation of cyclohexanol with hydrogen peroxide.
2. **Paracetamol:** Paracetamol is a widely used analgesic and antipyretic drug. It can be synthesized through the conventional route by the nitration of phenol, followed by the reduction of the nitro group to an amino group, and then acetylation. This process involves the use of hazardous chemicals and produces toxic waste. The green route for the synthesis of paracetamol involves the use of enzymatic methods or the use of microwave irradiation.

In summary, the synthesis of organic compounds such as adipic acid and paracetamol can be achieved through both conventional and green routes. The green route offers a more environmentally friendly alternative to the traditional chemical methods. It is important for students to understand the differences between these two routes and the benefits of using green chemistry in the synthesis of organic compounds.

LIST OF EXPERIMENTS

1. **Milgram Experiment:** A study on obedience to authority figures, conducted by Stanley Milgram in the 1960s.
2. **Stanford Prison Experiment:** A study of the psychological effects of becoming a prisoner or prison guard, conducted by Philip Zimbardo in 1971.
3. **Pavlov's Dogs:** A study on classical conditioning, conducted by Ivan Pavlov in the 1890s.

4. **Asch Conformity Experiment:** A study on conformity, conducted by Solomon Asch in the 1950s.
5. **Hawthorne Effect:** A study on the effect of changes in working conditions on productivity, conducted by Elton Mayo in the 1920s.
6. **Bobo Doll Experiment:** A study on the effects of observed aggression, conducted by Albert Bandura in the 1960s.
7. **Little Albert Experiment:** A study on classical conditioning of fear, conducted by John B. Watson and Rosalie Rayner in 1920.
8. **Marshmallow Test:** A study on delayed gratification, conducted by Walter Mischel in the 1960s.
9. **Bystander Effect:** A study on the diffusion of responsibility, conducted by John Darley and Bibb Latané in the 1960s.
10. **Rosenhan Experiment:** A study on the validity of psychiatric diagnosis, conducted by David Rosenhan in 1973.

1. Calibration of Analytical Equipment and Apparatus

Calibration is the process of verifying the accuracy of a measuring instrument by comparing it with a reference standard. This is important for analytical equipment and apparatus to ensure that the measurements obtained are accurate and reliable.

Here are some key points to consider when calibrating analytical equipment and apparatus:

- Calibration should be performed regularly to ensure the accuracy of the equipment.
- The reference standard used for calibration should be traceable to a national or international standard.
- The calibration process should be documented, including the date of calibration, the reference standard used, and the results obtained.
- Any deviations from the expected results should be investigated and corrective actions taken if necessary.
- The calibration process should be performed by trained personnel to ensure that it is carried out correctly.

By following these guidelines, you can ensure that your analytical equipment and apparatus are properly calibrated and provide accurate and reliable measurements.

2. Determination of Hardness of water sample by EDTA method

The hardness of water is due to the presence of calcium and magnesium salts in water. The determination of hardness of water by EDTA method is one of the three main methods for determination of hardness of water. EDTA stands for Ethylenediaminetetraacetic acid. This EDTA reagent can form EDTA-metal complexes by reacting with metal ions except for alkali metal ions.

The procedure for the calculation of hardness of water by EDTA titration is as follows:

1. Take a sample volume of 20ml (V ml).
2. Dilute 20ml of the sample in an Erlenmeyer flask to 40ml by adding 20ml of distilled water.
3. Add 1 mL of ammonia buffer to bring the pH to 10 ± 0.1 .
4. Add 1 or 2 drops of the indicator solution.
5. Add EDTA titrant to the sample with vigorous shaking until the wine-red color just turns blue.

The hardness of water can be calculated using the formula:

Hardness (EDTA), as mg/L = $(A * T * 1000) / S$

where A = mL of EDTA titrant used, T = Titer of EDTA titrant, mg CaCO₃ per mL of EDTA titrant, and S = mL of sample volume.

The result can be reported as “Total Hardness (EDTA) = _____ mg/L as CaCO₃” or as “Calcium Hardness (EDTA) = _____ mg/L as CaCO₃”.

3. Determination of Alkalinity of water sample

Alkalinity is a measure of the capacity of water to neutralize acids. It is determined by measuring the amount of acid needed to lower the pH of the water sample to a specific value. The alkalinity of water is primarily due to the presence of bicarbonate, carbonate, and hydroxide ions.

The procedure for determining the alkalinity of a water sample is as follows:

1. Collect a water sample in a clean container.
2. Measure the pH of the water sample using a pH meter.
3. Add a known amount of acid to the water sample while stirring.
4. Continue adding acid until the pH of the water sample reaches a specific value, usually 4.5.
5. Record the amount of acid added to the water sample.
6. Calculate the alkalinity of the water sample using the recorded data.

The alkalinity of the water sample is expressed in milligrams per liter (mg/L) of calcium carbonate (CaCO₃). A water sample with high alkalinity can neutralize more acid than a water sample with low alkalinity. Alkalinity is an important parameter in water treatment and environmental monitoring. It is used to assess the buffering capacity of water and its ability to resist changes in pH.

4. Determination of pH by titrimetric method

Titrimetric method is a common laboratory technique used to determine the pH of a solution. The process involves the following steps:

1. A solution of known concentration, called the titrant, is prepared. This is usually a strong acid or base.

2. The solution to be tested, called the analyte, is measured and placed in a container.
3. A few drops of an indicator are added to the analyte. The indicator is a substance that changes color depending on the pH of the solution.
4. The titrant is slowly added to the analyte while stirring. The indicator changes color when the pH of the solution reaches a certain value.
5. The volume of titrant added is recorded when the indicator changes color. This is called the endpoint.
6. The pH of the solution can be calculated using the volume of titrant added and the concentration of the titrant.

This method is accurate and precise, but it requires careful preparation and execution. It is commonly used in laboratories and in industrial processes where accurate pH measurements are required.

5. Determination of surface tension of given liquid

Surface tension is a property of liquids that arises due to the cohesive forces between the liquid molecules. It is defined as the force acting per unit length on the surface of the liquid. There are several methods to determine the surface tension of a given liquid, including:

1. **Capillary rise method:** This method involves measuring the height to which a liquid rises in a capillary tube due to the surface tension of the liquid. The surface tension is then calculated using the formula: $\text{Surface tension} = (\text{weight of liquid raised} * \text{radius of capillary tube}) / (\text{height of liquid rise} * 2 * \pi)$
2. **Maximum bubble pressure method:** This method involves measuring the maximum pressure required to blow a bubble of air through a liquid. The surface tension is then calculated using the formula: $\text{Surface tension} = (4 * \text{maximum pressure} * \text{radius of tube}) / (3 * \text{radius of bubble})$
3. **Drop weight method:** This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube. The surface tension is then calculated using the formula: $\text{Surface tension} = (\text{weight of droplet} * \text{acceleration due to gravity}) / (\pi * \text{radius of tube}^2)$
4. **Du Nouy ring method:** This method involves measuring the force required to detach a ring from the surface of a liquid. The surface tension is then calculated using the formula: $\text{Surface tension} = (\text{force required to detach ring}) / (2 * \pi * \text{radius of ring})$
5. **Wilhelmy plate method:** This method involves measuring the force required to detach a flat plate from the surface of a liquid. The surface tension is then calculated using the formula: $\text{Surface tension} = (\text{force required to detach plate}) / (\text{perimeter of plate})$

Each of these methods has its own advantages and limitations, and the choice of method depends on the specific requirements of the experiment. It is important to carefully calibrate the instruments and follow the experimental procedure to obtain accurate results.

6. Determination of Viscosity of a given liquid by viscometer

Viscosity is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate. A viscometer is an instrument used to measure the viscosity of a fluid. There are several types of viscometers, including capillary, rotational, and falling ball viscometers.

To determine the viscosity of a given liquid using a viscometer, the following steps can be followed:

1. Prepare the viscometer according to the manufacturer's instructions.
2. Fill the viscometer with the liquid to be tested.
3. Start the viscometer and record the time it takes for the liquid to flow through the viscometer.
4. Calculate the viscosity of the liquid using the appropriate formula for the type of viscometer being used.

It is important to note that the viscosity of a liquid can be affected by factors such as temperature and pressure. Therefore, it is important to conduct the test under controlled conditions to ensure accurate results.

7. Determination of the strength of Ferrous ammonium sulfate using external indicator

Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate. It is commonly used in the determination of the strength of oxidizing agents using an external indicator. Here are the steps involved in the process:

1. A solution of ferrous ammonium sulfate is prepared by dissolving a known amount of the salt in water.
2. The solution is then titrated against the oxidizing agent whose strength is to be determined.
3. An external indicator, such as potassium permanganate or diphenylamine, is used to determine the end point of the titration.
4. The end point is reached when the color of the indicator changes, indicating that all the ferrous ions have been oxidized to ferric ions.
5. The strength of the oxidizing agent can then be calculated using the volume of the solution used and the concentration of the ferrous ammonium sulfate solution.

This method is commonly used in analytical chemistry to determine the strength of oxidizing agents such as potassium permanganate, hydrogen peroxide, and

potassium dichromate. It is a simple and accurate method that can be easily performed in a laboratory setting.

8. Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is a primary standard substance that can be used to determine the strength of a solution.
2. An internal indicator is used to determine the end point of the titration.
3. The internal indicator used in this experiment is potassium iodide.
4. The reaction between potassium dichromate and potassium iodide produces iodine, which can be detected by its characteristic color.
5. The end point of the titration is reached when all the iodine has been consumed and the solution turns from a yellowish-brown color to a clear color.
6. The strength of the potassium dichromate solution can be calculated using the volume of the solution used and the known concentration of the potassium iodide solution.

9. Determination of available chlorine in bleaching powder

Bleaching powder, also known as calcium hypochlorite, is a chemical compound with the formula $\text{Ca}(\text{ClO})_2$. It is widely used as a disinfectant and bleaching agent. The available chlorine in bleaching powder refers to the amount of chlorine that can be released as a disinfectant when the powder is dissolved in water.

The available chlorine in bleaching powder can be determined by the following steps:

1. Weigh a sample of bleaching powder and dissolve it in water.
2. Add excess potassium iodide (KI) to the solution. The iodide ions will react with the available chlorine to produce iodine.
3. Titrate the iodine produced with a standard solution of sodium thiosulfate ($\text{Na}_2\text{S}_2\text{O}_3$) using starch as an indicator. The reaction between iodine and thiosulfate is as follows: $2\text{S}_2\text{O}_3^{2-} + \text{I}_2 \rightarrow \text{S}_4\text{O}_6^{2-} + 2\text{I}^-$
4. From the volume of thiosulfate solution used, the amount of iodine produced can be calculated. Since the reaction between iodine and thiosulfate is 1:1, the amount of available chlorine in the bleaching powder can be determined.

This method is known as iodometric titration and is commonly used to determine the available chlorine in bleaching powder. It is a simple, accurate, and reliable method for this purpose.

10. Determination of chloride content in water sample

1. Chloride content in water can be determined using a titration method known as the Mohr method.
2. In this method, a water sample is titrated with a solution of silver nitrate (AgNO_3) of known concentration.
3. Potassium chromate (K_2CrO_4) is used as an indicator, which forms a red precipitate of silver chromate (Ag_2CrO_4) when all the chloride ions have reacted with the silver ions.
4. The volume of silver nitrate solution used in the titration is noted and used to calculate the concentration of chloride ions in the water sample.
5. The calculation is done using the formula: $\text{Cl}^- (\text{mg/L}) = (\text{AgNO}_3 \text{ used (mL)} \times \text{AgNO}_3 \text{ normality} \times 35.45) / \text{sample volume (mL)}$.
6. This method is simple, accurate, and widely used for the determination of chloride content in water samples.
7. It is important to determine the chloride content in water as high levels of chloride can have adverse effects on human health, aquatic life, and the environment.
8. The World Health Organization (WHO) recommends a maximum limit of 250 mg/L for chloride in drinking water.
9. Regular monitoring of chloride levels in water sources is necessary to ensure the safety and quality of water for consumption and other uses.
10. Other methods for the determination of chloride content in water include ion chromatography, ion-selective electrodes, and spectrophotometry.

11. Preparation of Phenol formaldehyde (PF) resin

Phenol formaldehyde (PF) resin is a synthetic polymer that is obtained by the reaction of phenol with formaldehyde in the presence of an acid or a base catalyst. Here are the steps involved in the preparation of PF resin:

1. **Phenol and formaldehyde are mixed:** Phenol and formaldehyde are mixed in a molar ratio of 1:1.5 to 2.5 in a reaction vessel.
2. **Catalyst is added:** An acid or a base catalyst is added to the reaction mixture to initiate the polymerization reaction. Commonly used catalysts include hydrochloric acid, sulfuric acid, and sodium hydroxide.
3. **Heating:** The reaction mixture is heated to a temperature of 60-100°C for several hours to facilitate the polymerization reaction.
4. **Cooling:** The reaction mixture is cooled to room temperature, and the resulting PF resin is separated from the reaction mixture.
5. **Washing and drying:** The PF resin is washed with water to remove any unreacted phenol and formaldehyde, and then dried to obtain the final product.

PF resin is widely used in the production of laminates, adhesives, and molded

objects due to its high mechanical strength, heat resistance, and chemical resistance. It is also used as an insulating material in the electrical and electronics industries.

12. Preparation of Urea formaldehyde (UF) resin

Urea formaldehyde (UF) resin is a type of thermosetting polymer that is commonly used in the production of adhesives, finishes, and molded objects. Here are the steps involved in the preparation of UF resin:

1. **Reacting urea and formaldehyde:** The first step in the preparation of UF resin involves the reaction of urea and formaldehyde in an aqueous solution. The molar ratio of urea to formaldehyde is typically between 1:1.5 and 1:2.5. The reaction is carried out at a pH of 7-8 and at a temperature of 70-90°C.
2. **Condensation:** The next step involves the condensation of the reaction mixture. This is achieved by heating the mixture to a temperature of 90-100°C and maintaining it at this temperature for a period of time. The pH of the mixture is also adjusted to 4.5-5.5 during this step.
3. **Concentration:** The reaction mixture is then concentrated by evaporating the water. This is typically done under reduced pressure to prevent the resin from decomposing.
4. **Cooling and storage:** The concentrated resin is then cooled and stored for future use. The resin can be stored in a cool, dry place for several months.

These are the basic steps involved in the preparation of UF resin. The exact conditions and parameters may vary depending on the specific application and desired properties of the final product. It is important to carefully control the reaction conditions to ensure the production of a high-quality resin.

13. Preparation of Adipic acid / Paracetamol

Adipic acid and Paracetamol are two different compounds with different chemical structures and properties. Here is the information on the preparation of each compound:

Adipic Acid:

- Adipic acid is an organic compound with the formula $(CH_2)_4(COOH)_2$.
- It is a white crystalline powder that is soluble in water and alcohol.
- Adipic acid is mainly used in the production of nylon.
- It can be prepared by the oxidation of cyclohexanol or cyclohexanone with nitric acid.
- Another method of preparation involves the hydrolysis of cyclohexanone oxime with sulfuric acid.

Paracetamol:

- Paracetamol, also known as acetaminophen, is an analgesic and antipyretic drug.
- It is commonly used to relieve pain and reduce fever.
- Paracetamol can be prepared by the acetylation of p-aminophenol with acetic anhydride.
- Another method of preparation involves the nitration of phenol followed by reduction to p-aminophenol and subsequent acetylation.

14. Determination of Cell Conductance of a solution

Cell conductance is a measure of a solution's ability to conduct an electric current. It is determined by the concentration and mobility of ions in the solution. Here are the steps to determine the cell conductance of a solution:

1. Prepare the solution whose conductance is to be measured.
2. Clean the conductivity cell with distilled water and dry it.
3. Fill the cell with the solution and ensure that the electrodes are completely immersed.
4. Connect the cell to a conductivity meter and switch it on.
5. Allow the reading to stabilize and record the conductance value displayed on the meter.
6. Repeat the measurement with different concentrations of the solution to study the variation of conductance with concentration.

It is important to calibrate the conductivity meter with a standard solution of known conductance before taking measurements. The cell constant of the conductivity cell should also be determined to ensure accurate results.

15. Determination of Rate constant of hydrolysis of esters

1. The rate constant of hydrolysis of esters can be determined by measuring the rate of reaction at different concentrations of the reactants.
2. The reaction between an ester and water to form a carboxylic acid and an alcohol is known as hydrolysis.
3. The rate of hydrolysis of an ester can be determined by measuring the change in concentration of one of the reactants or products over time.
4. The rate law for the hydrolysis of an ester can be written as: $\text{rate} = k[\text{ester}]$, where k is the rate constant.
5. The rate constant, k , can be determined by measuring the initial rate of the reaction at different initial concentrations of the ester and water.
6. A graph of the initial rate versus the initial concentration of the ester can be plotted, and the slope of the line will give the value of the rate constant, k .
7. The units of the rate constant will depend on the order of the reaction with respect to the ester and water.

8. The order of the reaction can be determined by varying the initial concentration of one of the reactants while keeping the other constant and observing the effect on the initial rate of the reaction.
9. The hydrolysis of esters is typically catalyzed by acids or bases, and the rate constant will depend on the pH of the solution.
10. The temperature of the reaction will also affect the rate constant, with higher temperatures leading to faster reaction rates.

16. Element detection and identification of functional groups in organic compounds

- Element detection in organic compounds involves the determination of the presence of certain elements such as carbon, hydrogen, nitrogen, sulfur, and halogens.
- Various tests can be performed to detect the presence of these elements. For example, the Lassaigne's test can be used to detect the presence of nitrogen, sulfur, and halogens.
- Identification of functional groups in organic compounds involves the determination of the specific functional groups present in the compound.
- Various tests can be performed to identify the functional groups present in the compound. For example, the bromine water test can be used to identify the presence of a carbon-carbon double bond, and the Tollens' test can be used to identify the presence of an aldehyde functional group.
- The identification of functional groups is important in the determination of the chemical properties and reactivity of the organic compound.
- Knowledge of the functional groups present in an organic compound can also aid in the determination of the compound's structure.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose 10 experiments from the list provided.
- The instructor also has the option to change any two of the experiments from the list.
- This allows the instructor to tailor the experiments to the needs and interests of the students.
- It is important for students to carefully read and understand the instructions provided by the instructor.
- Students should also be prepared for any changes to the list of experiments.

Course Outcomes:

1. Understand the fundamental concepts and principles of the subject matter.
2. Develop the ability to apply the knowledge and skills acquired in the course to solve problems and make informed decisions.

3. Demonstrate the ability to communicate effectively, both orally and in writing.
4. Develop critical thinking and analytical skills.
5. Demonstrate the ability to work effectively in a team.
6. Develop an appreciation for the ethical and social implications of the subject matter.
7. Demonstrate the ability to engage in lifelong learning and professional development.

Upon completion of the course the student should be able to:

1. Demonstrate a comprehensive understanding of the subject matter.
2. Apply the knowledge and skills acquired in the course to real-world scenarios.
3. Analyze and evaluate information critically and effectively.
4. Communicate effectively in both written and oral forms.
5. Work collaboratively in a team environment.
6. Demonstrate ethical and professional behavior.
7. Engage in lifelong learning and professional development.
8. Utilize technology effectively in the learning process.
9. Demonstrate problem-solving and decision-making skills.
10. Think creatively and critically to generate innovative solutions.

Course Outcomes Bloom's

Course outcomes are statements that describe the knowledge, skills, and abilities that students should possess upon completion of a course. These outcomes are typically aligned with the course objectives and are used to assess student learning.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. The taxonomy consists of six levels, with the lower levels representing basic knowledge and comprehension, and the higher levels representing more complex cognitive processes such as analysis, synthesis, and evaluation.

The six levels of Bloom's Taxonomy are:

1. **Remembering:** recalling previously learned information.
2. **Understanding:** comprehending the meaning of the information.
3. **Applying:** using the information in a new situation.
4. **Analyzing:** breaking down the information into parts to understand its structure.
5. **Evaluating:** making judgments about the value of the information.
6. **Creating:** combining information to form a new whole.

By aligning course outcomes with Bloom's Taxonomy, instructors can ensure that their course is designed to promote higher-order thinking and that assess-

ments are appropriately challenging for students. This can help students develop the skills and knowledge necessary to succeed in their academic and professional careers.

Level

- A level is a tool used to determine if a surface is horizontal (level) or vertical (plumb).
- Levels can be used in construction, carpentry, surveying, and many other applications.
- There are several types of levels, including spirit levels, laser levels, and water levels.
- Spirit levels use a liquid-filled vial with an air bubble to indicate levelness. The bubble will be centered between two lines when the surface is level.
- Laser levels project a laser beam to create a straight line, which can be used to determine levelness over a longer distance.
- Water levels use the principle that water will always find its own level. A clear tube filled with water can be used to determine levelness between two points.
- Levels can vary in size and accuracy, with larger and more precise levels being used for more demanding applications.
- It is important to use a level when installing or building structures to ensure that they are straight and level, which can affect their stability and appearance.

CO-1 Get an understanding of the use of different analytical instruments. K3

Analytical instruments are used to measure physical, chemical, and biological properties of substances. These instruments are used in a variety of fields, including chemistry, biology, physics, and engineering. Some common analytical instruments include:

1. **Spectrophotometers:** These instruments measure the absorption or transmission of light by a sample. They are commonly used to determine the concentration of a substance in a solution.
2. **Chromatographs:** These instruments separate mixtures into their individual components. They are commonly used to identify and quantify the components of a mixture.
3. **Mass spectrometers:** These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify the chemical composition of a sample.
4. **Electrochemical instruments:** These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.

5. **Microscopes:** These instruments magnify small objects. They are commonly used to observe the structure of cells and tissues.

Understanding the use of these instruments is important for conducting accurate and precise measurements in scientific research and industry. By using these instruments, scientists and engineers can gain a better understanding of the properties of substances and how they interact with each other. This knowledge can be used to develop new products, improve existing processes, and advance scientific understanding.

CO-2 Measure the molecular / system properties such as surface tension

Surface tension is a property of liquids that arises due to the cohesive forces between the molecules at the surface of the liquid. These forces cause the liquid to behave as if it has an elastic skin on its surface. Surface tension is typically measured in units of force per unit length, such as Newtons per meter (N/m).

There are several methods for measuring surface tension, including:

1. The maximum bubble pressure method: This method involves measuring the maximum pressure required to force a gas bubble through a liquid.
2. The drop weight method: This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube.
3. The Wilhelmy plate method: This method involves measuring the force required to detach a thin, flat plate from the surface of a liquid.
4. The Du Nouy ring method: This method involves measuring the force required to detach a thin, circular ring from the surface of a liquid.

Each of these methods has its own advantages and disadvantages, and the choice of method will depend on the specific requirements of the measurement. It is important to carefully calibrate and control the experimental conditions when measuring surface tension to ensure accurate and reliable results.

Viscosity, Conductance of Solution, Chloride and Iron Content in the Water

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.
- The viscosity of formation water is a function of pressure, temperature, and dissolved solids.
- In general, brine viscosity increases with increasing pressure, increasing salinity, and decreasing temperature.
- Dissolved gas in the formation water at reservoir conditions generally results in a negligible effect on water viscosity.

Conductance of Solution

- Conductivity is a measure of a solution's ability to conduct an electric current.
- High-quality deionized water has a conductivity of about 0.05 S/cm at 25 °C, typical drinking water is in the range of 200–800 S/cm, while seawater is about 50 mS/cm (or 0.05 S/cm).
- The more chloride ions present in the water, the higher the conductivity.
- In general, the conductivity of waterways in the United States varies from 50 to 1500 µmhos/cm, and inland freshwater lake studies reveal a conductivity of 150 to 500 µmhos/cm.

Chloride Content in Water

- Chloride ions are approximately 1.9 percent of the mass of seawater.

Iron Content in Water

- Iron content in water was not found in the search results.

K3

- K3 is a term that can refer to several different things, depending on the context.
- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a visa category for spouses of U.S. citizens.
- In mathematics, K3 could refer to a K3 surface, which is a complex algebraic surface with certain properties.
- In music, K3 is a Belgian-Dutch girl group.

CO-3 Measure the hardness and alkalinity of the water. K3

Hardness and alkalinity are two important parameters of water quality. Hardness refers to the concentration of calcium and magnesium ions in water, while alkalinity refers to the ability of water to neutralize acids.

1. **Measuring Hardness:** Hardness can be measured using a titration method, where a solution of known concentration is added to a water sample until a color change occurs. The amount of solution required to reach the endpoint is used to calculate the hardness of the water. Another method is to use a hardness test strip, which changes color based on the hardness of the water.
2. **Measuring Alkalinity:** Alkalinity can also be measured using a titration method, where a solution of known concentration is added to a water sample until a color change occurs. The amount of solution required to reach the endpoint is used to calculate the alkalinity of the water. Another method is to use an alkalinity test kit, which contains a reagent that changes color based on the alkalinity of the water.

It is important to measure the hardness and alkalinity of water to ensure that it is suitable for its intended use. For example, water with high hardness can cause scaling in pipes and appliances, while water with low alkalinity can be corrosive.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an organic compound with the chemical formula C_6H_5OH . It is also known as carbolic acid, hydroxybenzene, or phenic acid. Phenol is a white crystalline solid at room temperature and is soluble in water, alcohol, and ether. It has a characteristic odor and is used in the production of many chemicals, including plastics, resins, and pharmaceuticals.

There are several methods for the preparation of phenol, including:

1. **From Cumene:** Cumene is oxidized in the presence of air to produce cumene hydroperoxide, which is then cleaved in the presence of an acid catalyst to produce phenol and acetone.
2. **From Chlorobenzene:** Chlorobenzene is heated with an aqueous solution of sodium hydroxide at high temperature and pressure to produce phenol and sodium chloride.
3. **From Benzene:** Benzene is first converted to nitrobenzene by nitration, then reduced to aniline by hydrogenation. Aniline is then diazotized and treated with water to produce phenol.
4. **From Benzene Sulphonic Acid:** Benzene is sulfonated to produce benzene sulfonic acid, which is then fused with an alkali to produce phenol and a sulfite salt.

These are some of the fundamental concepts of the preparation of phenol. It is important to understand the chemical reactions and processes involved in the production of this important compound.

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

- Formaldehyde is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.
- Urea-formaldehyde (UF) is a nontransparent thermosetting resin or polymer produced from urea and formaldehyde.
- UF is synthesized by the “alkali-acid-alkali” method and often shows the appearance of crystal domains .
- UF resins are used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- Melamine, polyvinyl alcohol, and adipic acid dihydrazide can be added to UF resin as modifiers for reducing the free formaldehyde.

K3

- K3 is a term that can refer to several different things, depending on the context in which it is used.
- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a visa category for spouses of U.S. citizens.
- In mathematics, K3 could refer to the complete graph on three vertices.
- In music, K3 is a Belgian-Dutch girl group.
- Without more context, it is difficult to determine which specific meaning of K3 is being referred to.

CO-5 Estimate the rate constant of reaction. K3

The rate constant of a reaction, denoted by the symbol k , is a measure of the speed of a chemical reaction. It is defined as the proportionality constant between the rate of the reaction and the concentrations of the reactants raised to their respective powers, as given by the rate law of the reaction.

To estimate the rate constant of a reaction, the following steps can be followed:

1. Determine the rate law of the reaction by conducting experiments and analyzing the data to find the order of the reaction with respect to each reactant.
2. Measure the rate of the reaction at different concentrations of the reactants and at a constant temperature.
3. Plot the data on a graph with the rate of the reaction on the y-axis and the concentration of the reactants on the x-axis.
4. Use the slope of the line to calculate the rate constant, k , using the equation of the rate law.

It is important to note that the rate constant is temperature-dependent, and its value will change with changes in temperature. The Arrhenius equation can be used to calculate the rate constant at different temperatures.

Page 33 of 40

- Page 33 is the thirty-third page of a forty-page document.
- It is located after page 32 and before page 34.
- The content of page 33 will depend on the specific document it is a part of.
- It is important to read and understand the content of page 33 in the context of the entire document.
- Page 33 may contain text, images, tables, or other types of information.
- It is important to review page 33 carefully to ensure that the information is accurate and relevant.

- If studying for an exam, it may be helpful to take notes or highlight key information on page 33 to aid in retention and understanding.

ENGINEERING GRAPHICS LAB KCS

Engineering Graphics is a subject that is taught to engineering students in their first year. It is a basic course that helps students to understand the fundamentals of technical drawing and computer-aided design (CAD). The subject is also known as Engineering Drawing or Technical Drawing.

The Engineering Graphics Lab is a practical course that complements the theoretical concepts taught in the classroom. In the lab, students get hands-on experience in creating technical drawings using various tools and techniques.

Some of the key topics covered in the Engineering Graphics Lab include:

1. **Introduction to Engineering Graphics:** In this topic, students are introduced to the basic concepts of engineering graphics, including the use of drawing instruments, lettering, and dimensioning.
2. **Orthographic Projection:** This topic covers the principles of orthographic projection, including the creation of multi-view drawings and the use of auxiliary views.
3. **Isometric Projection:** In this topic, students learn how to create isometric drawings, which are used to represent 3D objects in 2D.
4. **Sectional Views:** This topic covers the creation of sectional views, which are used to show the internal details of an object.
5. **Computer-Aided Design (CAD):** In this topic, students are introduced to the basics of CAD software and learn how to create technical drawings using a computer.

The Engineering Graphics Lab is an essential course for engineering students, as it helps them to develop the skills needed to create technical drawings and communicate their ideas effectively. It is a practical subject that provides students with hands-on experience in using various tools and techniques to create technical drawings.

Course Objectives:

- To provide students with a comprehensive understanding of the subject matter.
- To develop critical thinking and analytical skills.
- To enhance problem-solving abilities.
- To encourage independent learning and self-motivation.
- To promote effective communication and collaboration.

- To prepare students for further academic or professional pursuits in the field.
- To foster an appreciation for the subject and its real-world applications.
- To instill ethical and responsible behavior in the practice of the subject.

Fundamental Concepts of Analytical Instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are essential tools in scientific research, medical diagnosis, and industrial processes. Here are some fundamental concepts to understand about analytical instruments:

1. **Accuracy and Precision:** Accuracy refers to how close a measurement is to the true value, while precision refers to how close repeated measurements are to each other. Analytical instruments must be both accurate and precise to provide reliable results.
2. **Sensitivity:** Sensitivity refers to the ability of an instrument to detect small changes in the property being measured. A highly sensitive instrument can detect even minute changes, making it useful for detecting trace amounts of substances.
3. **Selectivity:** Selectivity refers to the ability of an instrument to distinguish between different substances. A highly selective instrument can accurately identify and measure a specific substance, even in the presence of other substances.
4. **Calibration:** Calibration is the process of adjusting an instrument to ensure that it provides accurate and precise measurements. This is typically done by comparing the instrument's readings to known standards.
5. **Signal-to-Noise Ratio:** The signal-to-noise ratio (SNR) is a measure of the quality of a measurement. A high SNR indicates that the signal being measured is much stronger than the background noise, resulting in a more reliable measurement.

These are just a few of the fundamental concepts to understand about analytical instruments. By understanding these concepts, students can better appreciate the capabilities and limitations of these important tools.

Analysis of Chloride Content, Hardness, and Alkalinity

1. **Chloride Content:** Chloride content refers to the amount of chloride ions present in a water sample. Chloride ions are negatively charged particles that can be found in many types of water, including seawater, groundwater, and surface water. The analysis of chloride content is important because

high levels of chloride can indicate contamination from sources such as sewage, industrial waste, or road salt.

2. **Hardness:** Hardness refers to the concentration of calcium and magnesium ions in water. These ions can cause problems in industrial and domestic settings by forming deposits on pipes and appliances. The analysis of water hardness is important for determining the need for water softening or other treatment methods.
3. **Alkalinity:** Alkalinity refers to the ability of water to neutralize acids. It is a measure of the buffering capacity of water, or its ability to resist changes in pH. The analysis of alkalinity is important for understanding the chemical balance of water and its suitability for various uses.

In summary, the analysis of chloride content, hardness, and alkalinity is important for understanding the chemical properties of water and its suitability for various uses. These analyses can provide valuable information for water treatment and management.

Water

- Water is a substance composed of the chemical elements hydrogen and oxygen and existing in gaseous, liquid, and solid states.
- It is one of the most plentiful and essential of compounds.
- A tasteless and odourless liquid at room temperature, it has the important ability to dissolve many other substances.
- Water is a tiny molecule. It consists of three atoms: two of hydrogen and one of oxygen.
- Water molecules cling to each other because of a force called hydrogen bonding.
- Water is a shape-shifter. It exists in three states on Earth: liquid, gas, and solid.
- Water is the liquid that makes life on Earth possible.
- As water cycles from the air to the land to the sea and back again, water shapes our planet and nearly every aspect of our lives.
- All living things, from tiny cyanobacteria to giant blue whales, need water to survive.
- Water is an inorganic compound with the chemical formula H_2O .
- It is a transparent, tasteless, odorless, and nearly colorless chemical substance.
- It is the main constituent of Earth's hydrosphere and the fluids of all known living organisms (in which it acts as a solvent).

Measure of pH, Surface Tension and Viscosity

1. **pH:** pH is a measure of the acidity or basicity of a solution. It is defined as the negative logarithm of the hydrogen ion concentration in a solution. The pH scale ranges from 0 to 14, with 7 being neutral. A pH less than 7 indicates an acidic solution, while a pH greater than 7 indicates a basic solution.
2. **Surface Tension:** Surface tension is the property of a liquid that allows it to resist an external force. It is caused by the cohesive forces between the molecules of the liquid. Surface tension can be measured using a number of methods, including the drop weight method, the maximum bubble pressure method, and the Wilhelmy plate method.
3. **Viscosity:** Viscosity is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate in a fluid. Viscosity can be measured using a number of methods, including the capillary tube method, the falling ball method, and the rotational viscometer method.

These three properties are important in many fields, including chemistry, physics, and engineering. Understanding how to measure and manipulate these properties can be useful in a variety of applications.

Liquid

- A liquid is one of the four fundamental states of matter, along with solid, gas, and plasma.
- It is characterized by its ability to conform to the shape of its container, while retaining a constant volume.
- Liquids have a definite volume, but no definite shape.
- The molecules in a liquid are in constant motion and are loosely packed, allowing them to slide past one another.
- The attractive forces between the molecules in a liquid are strong enough to keep the molecules close together, but not strong enough to keep them in a fixed position.
- The temperature at which a substance changes from a solid to a liquid is called its melting point, and the temperature at which it changes from a liquid to a gas is called its boiling point.
- Liquids can be poured and spilled, and they can flow and take the shape of their container.
- Some common examples of liquids include water, milk, and oil.
- Liquids can be mixed with other liquids to form solutions, such as when sugar is dissolved in water to make sugar water.
- The study of liquids and their properties is called fluid mechanics.

Preparation of Different Resins

Resins are solid or highly viscous substances that are typically convertible into polymers. They are used in a variety of applications, including adhesives, coatings, and plastics. Here are some points to consider when preparing different types of resins:

1. **Polyester Resins:** These are typically prepared by the polycondensation of a polyol with a dibasic acid or anhydride. The most common type of polyester resin is unsaturated polyester resin, which is prepared by reacting a dibasic acid with a diol and a reactive monomer, such as styrene.
2. **Epoxy Resins:** These are typically prepared by the reaction of bisphenol A and epichlorohydrin. The resulting resin is then cured with a hardener, such as an amine or anhydride, to form a crosslinked polymer.
3. **Phenolic Resins:** These are typically prepared by the reaction of phenol with formaldehyde in the presence of an acid or base catalyst. The resulting resin can be cured with heat to form a crosslinked polymer.
4. **Silicone Resins:** These are typically prepared by the hydrolysis and condensation of silanes or siloxanes. The resulting resin can be cured with heat or a catalyst to form a crosslinked polymer.

It is important to carefully control the reaction conditions, such as temperature, time, and catalyst concentration, to ensure the desired properties of the final resin product. Additionally, the choice of starting materials and curing agents can greatly affect the properties of the final resin.

Synthesis of Adipic Acid

Adipic acid is an important organic compound that is used in the production of nylon, polyurethane, and other industrial products. Here are some key points to understand about the synthesis of adipic acid:

1. Adipic acid is a dicarboxylic acid with the chemical formula $C_6H_{10}O_4$.
2. It can be synthesized from a variety of starting materials, including cyclohexanol, cyclohexanone, and benzene.
3. One common method for synthesizing adipic acid is through the oxidation of cyclohexanol or cyclohexanone using nitric acid.
4. Another method involves the oxidation of benzene using a mixture of nitric and sulfuric acids.
5. The resulting adipic acid can be purified through recrystallization or other separation techniques.

These are some of the key points to understand about the synthesis of adipic acid. It is an important industrial chemical with a variety of uses, and its synthesis can be achieved through several different methods.

Acid and Paracetamol

- Mefenamic Acid + Paracetamol is a combination of two medicines: Mefenamic Acid and Paracetamol. It works by blocking the release of certain chemical messengers that cause fever, pain and inflammation (redness and swelling).
- Mefenamic Acid+Paracetamol is used for pain relief and fever. It relieves pain and inflammation in conditions like rheumatoid arthritis, osteoarthritis and menstrual pain. It is also used to treat mild to moderate pain.
- Paracetamol may be made by acetylation of para-aminophenol (obtained by reduction of para-nitrophenol) with acetic acid or acetic anhydride.
- Paracetamol is used as an analgesic and antipyretic drug. It is the preferred alternative analgesic-antipyretic to aspirin (acetylsalicylic acid), particularly in patients with coagulation disorders, individuals with a history of peptic ulcer or who cannot tolerate aspirin, as well as in children.
- Paracetamol (Panadol, Calpol, Alvedon) is an analgesic and antipyretic drug that is used to temporarily relieve mild-to-moderate pain and fever. It is commonly included as an ingredient in cold and flu medications and is also used on its own. Paracetamol is exactly the same drug as acetaminophen (Tylenol).